Microsoft ADC Cybersecurity Skilling Program

Week 11 Assignment 22

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Azure Monitor, Microsoft Defender for cloud, Enable Just-In Time Access in VMs, Microsoft Sentinel

Introduction

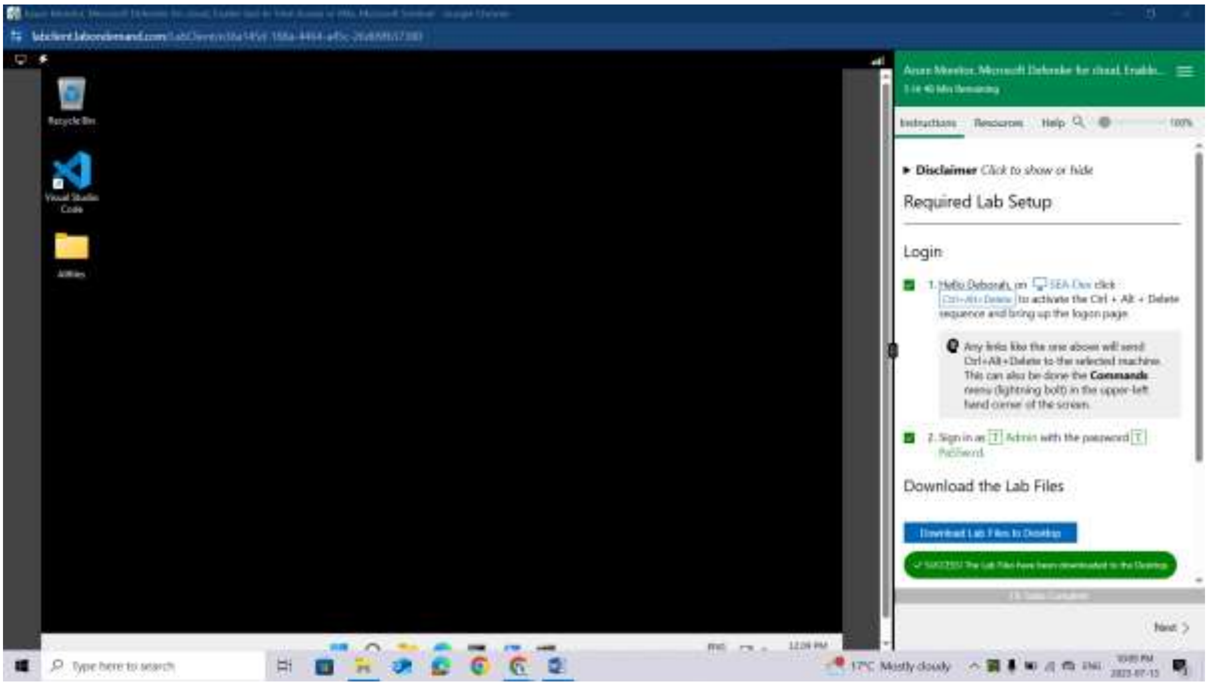
In this lab, I will explore key Azure security and monitoring tools to enhance visibility, protection, and incident response capabilities across cloud resources. The lab begins by working with **Azure Monitor** to collect, analyze, and act on telemetry data for performance and health insights. I will then configure **Microsoft Defender for Cloud** to assess the security posture of my environment and apply recommended hardening actions. To reduce the attack surface on virtual machines, I will enable **Just-In-Time (JIT) VM access**, allowing controlled access to VMs only when needed. Lastly, I will use **Microsoft Sentinel**, a cloud-native SIEM and SOAR solution, to detect, investigate, and respond to security threats using analytics, workbooks, and automation rules. This comprehensive approach will demonstrate how these integrated tools provide end-to-end security management in Azure.

Tasks Completed

Hello Deborah, on SEA-Dev click Ctrl+Alt+Delete to activate the Ctrl + Alt + Delete sequence and bring up the logon page.

Sign in as Admin with the password Pa55w.rd.

Download the Lab Files



Lab 08: Create a Log Analytics Workspace, Azure Storage Account, and Data Collection Rule (DCR)

Student lab manual

Lab scenario

As an Azure Security Engineer for a financial technology company, you are tasked with enhancing monitoring and security visibility across all Azure virtual machines (VMs) used for processing financial transactions and managing sensitive customer data. The security team requires detailed logs and performance metrics from these VMs to detect potential threats and optimize system performance. The Chief Information Security Officer (CISO) has asked you to implement a solution that collects security events, system logs, and performance counters. You have been assigned to configure the Azure Monitor Agent (AMA) along with Data Collection Rules (DCRs) to centralize log collection and performance monitoring.

*For all the resources in this lab, we are using the **East US** region. Verify with your instructor this is the region to use for class.*

Lab objectives

In this lab, you will complete the following exercises:

- Exercise 1: Deploy an Azure virtual machine
- Exercise 2: Create a Log Analytics workspace
- Exercise 3: Create an Azure storage account
- Exercise 4: Create a data colllection rule

Instructions

Exercise 1: Deploy an Azure virtual machine

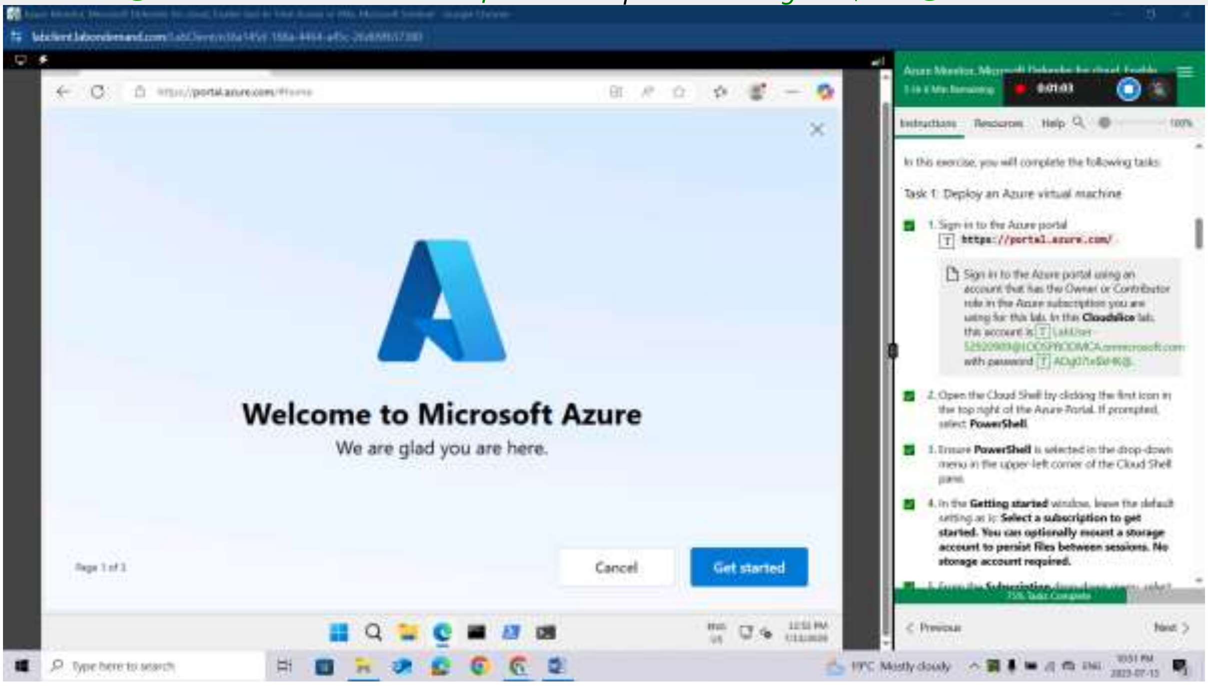
Exercise timing: 10 minutes

In this exercise, you will complete the following tasks:

Task 1: Deploy an Azure virtual machine

Sign-in to the Azure portal `https://portal.azure.com/`.

*Sign in to the Azure portal using an account that has the Owner or Contributor role in the Azure subscription you are using for this lab. In this **Cloudslice** lab, this account is LabUser-52920909@LODSPRODMCA.onmicrosoft.com with password ADg07!e$kHK@.*



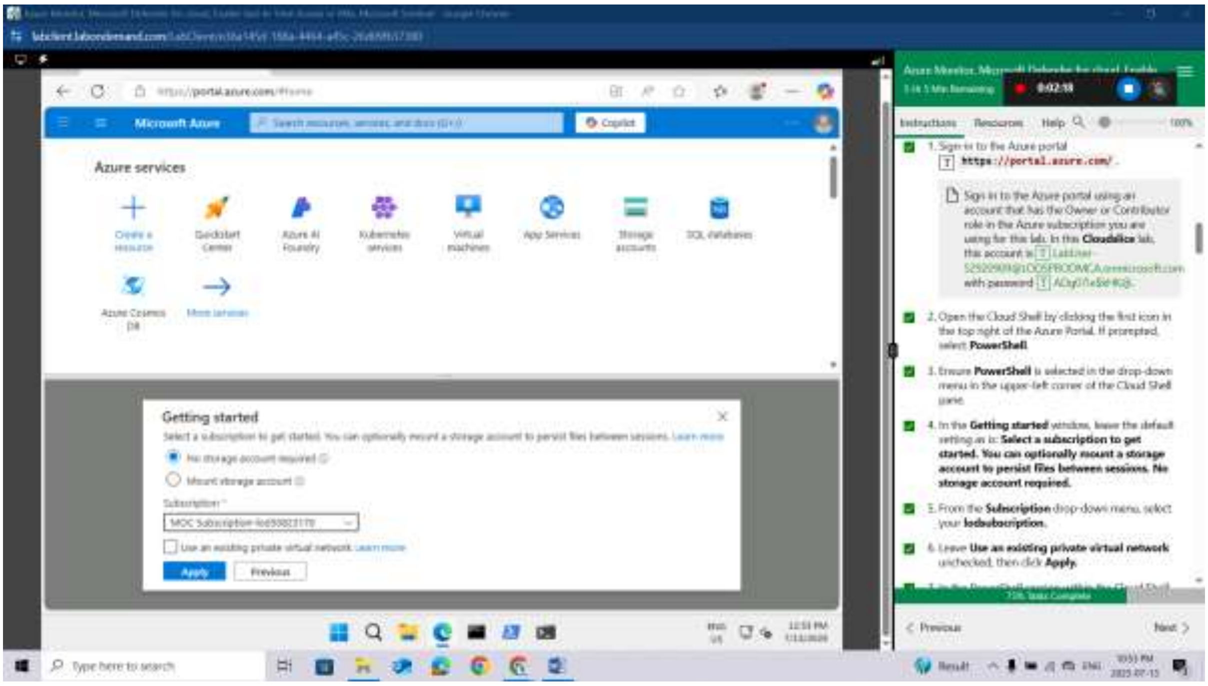
Open the Cloud Shell by clicking the first icon in the top right of the Azure Portal. If prompted, select **PowerShell**.

Ensure **PowerShell** is selected in the drop-down menu in the upper-left corner of the Cloud Shell pane

In the **Getting started** window, leave the default setting as is: **Select a subscription to get started. You can optionally mount a storage account to persist files between sessions. No storage account required.**

From the **Subscription** drop-down menu, select your **lodsubscription.**

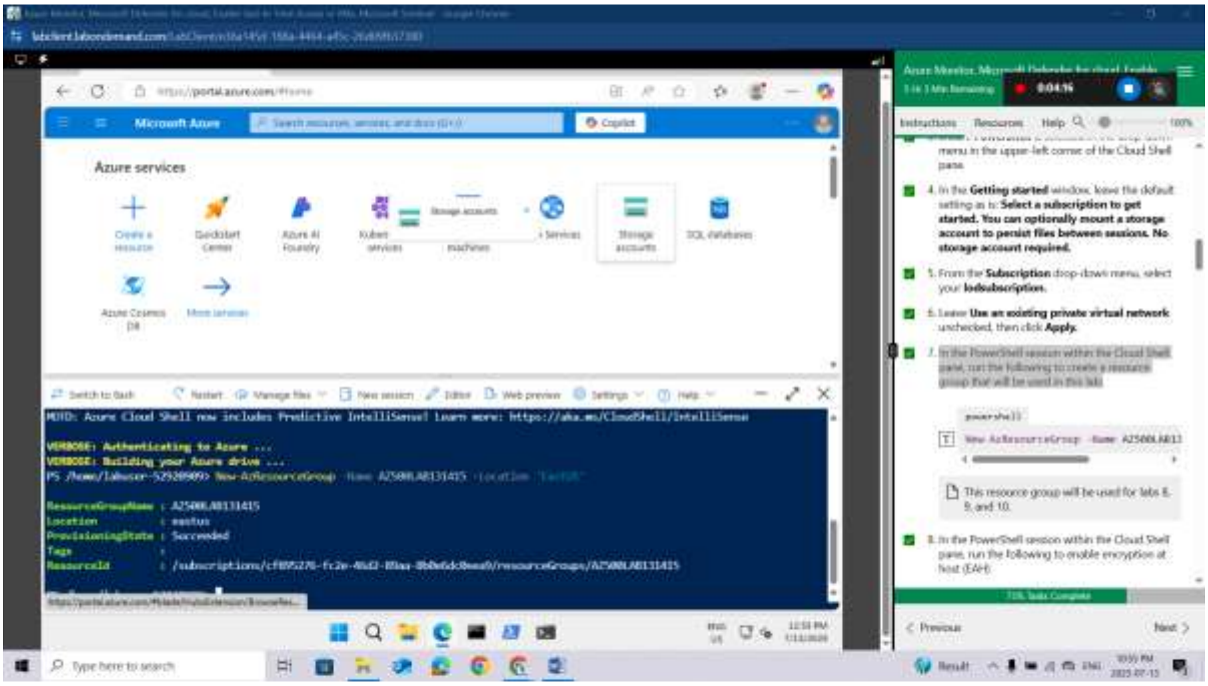
Leave **Use an existing private virtual network** unchecked, then click **Apply.**



In the PowerShell session within the Cloud Shell pane, run the following to create a resource group that will be used in this lab:

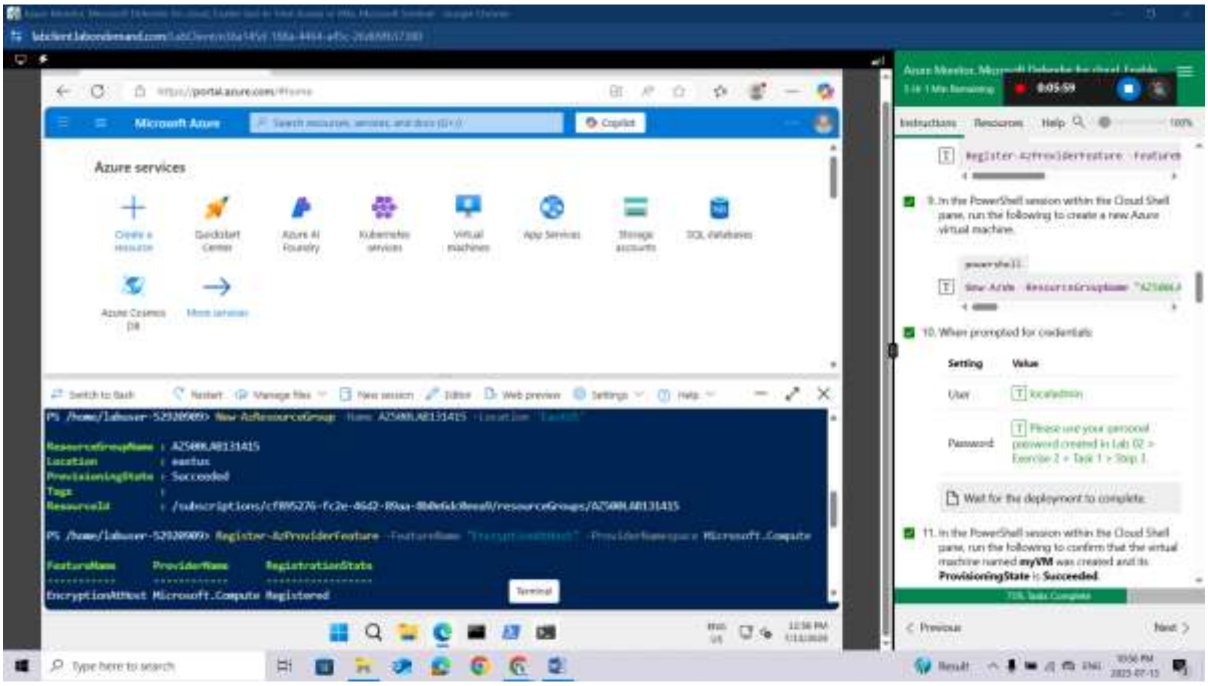
```powershell
New-AzResourceGroup -Name AZ500LAB131415 -Location 'EastUS'
```

This resource group will be used for labs 8, 9, and 10.



In the PowerShell session within the Cloud Shell pane, run the following to enable encryption at host (EAH)

```powershell
Register-AzProviderFeature -FeatureName "EncryptionAtHost" -ProviderNamespace
Microsoft.Compute
```



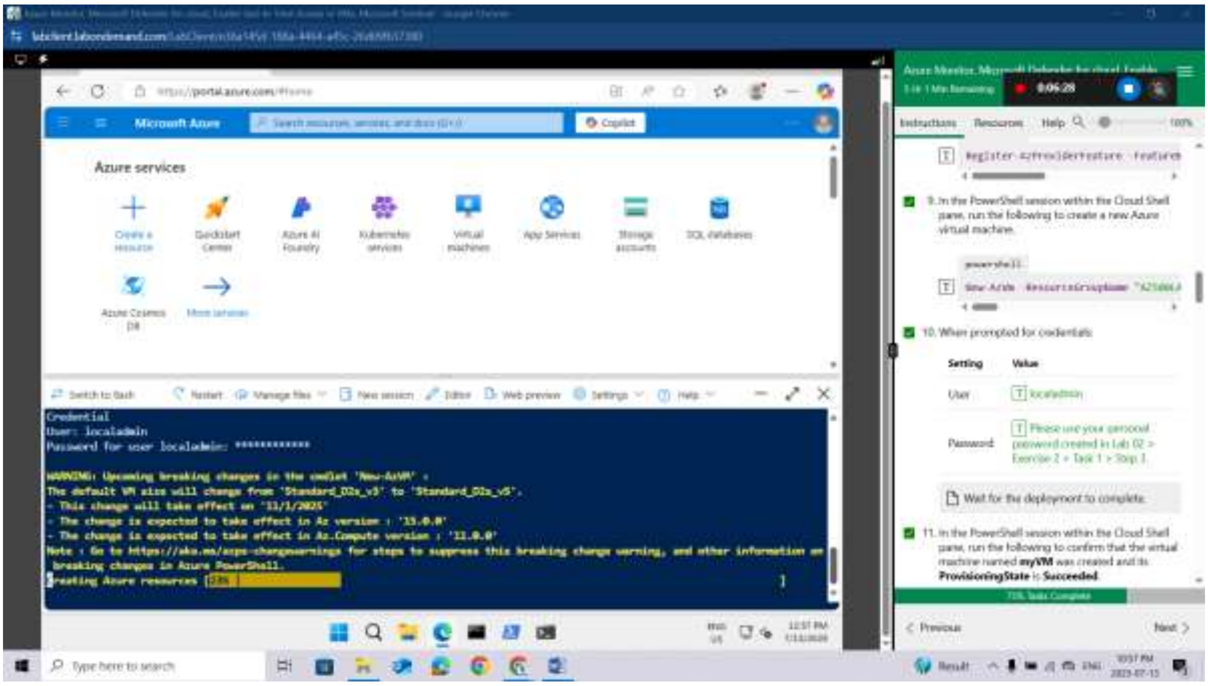
In the PowerShell session within the Cloud Shell pane, run the following to create a new Azure virtual machine.

```powershell
New-AzVm -ResourceGroupName "AZ500LAB131415" -Name "myVM" -Location 'EastUS' -VirtualNetworkName "myVnet" -SubnetName "mySubnet" -SecurityGroupName "myNetworkSecurityGroup" -PublicIpAddressName "myPublicIpAddress" -PublicIpSku Standard -OpenPorts 80,3389 -Size Standard_D2s_v3
```

When prompted for credentials:

Setting	Value
User	localadmin
Password	Please use your personal password created in Lab 02 > Exercise 2 > Task 1 > Step 3.

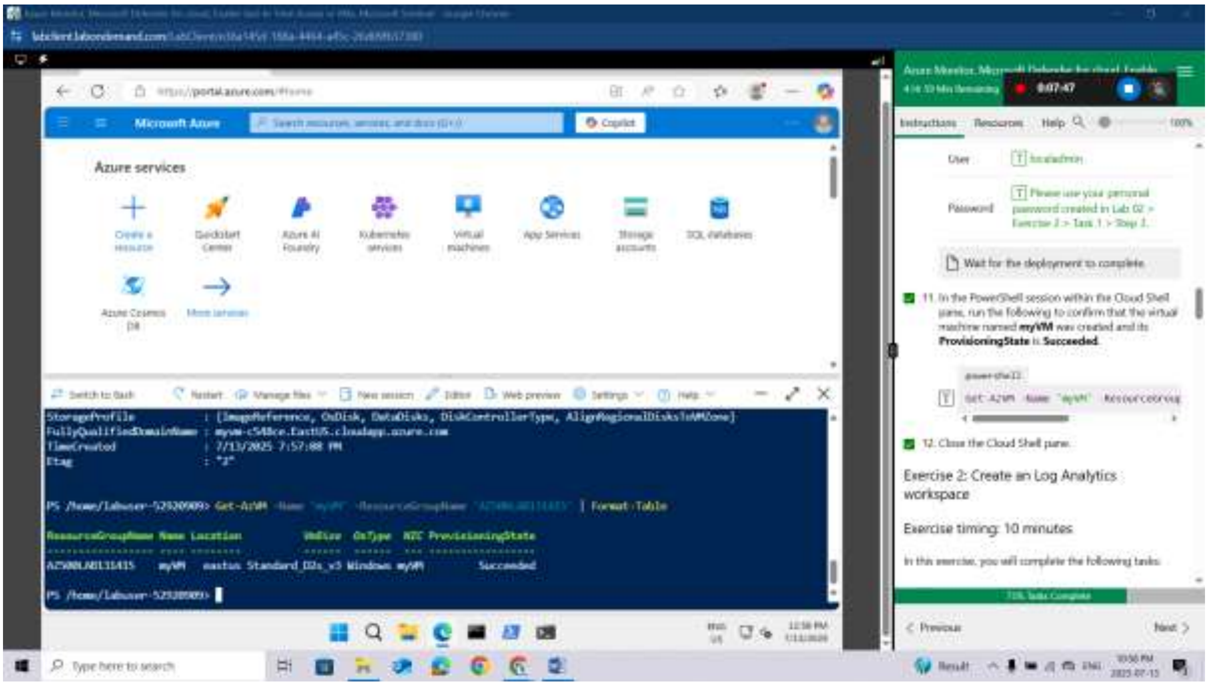
Wait for the deployment to complete.



In the PowerShell session within the Cloud Shell pane, run the following to confirm that the virtual machine named **myVM** was created and its **ProvisioningState** is **Succeeded**.

```powershell
Get-AzVM -Name 'myVM' -ResourceGroupName 'AZ500LAB131415' | Format-Table
```

Close the Cloud Shell pane.



Exercise 2: Create an Log Analytics workspace

Exercise timing: 10 minutes

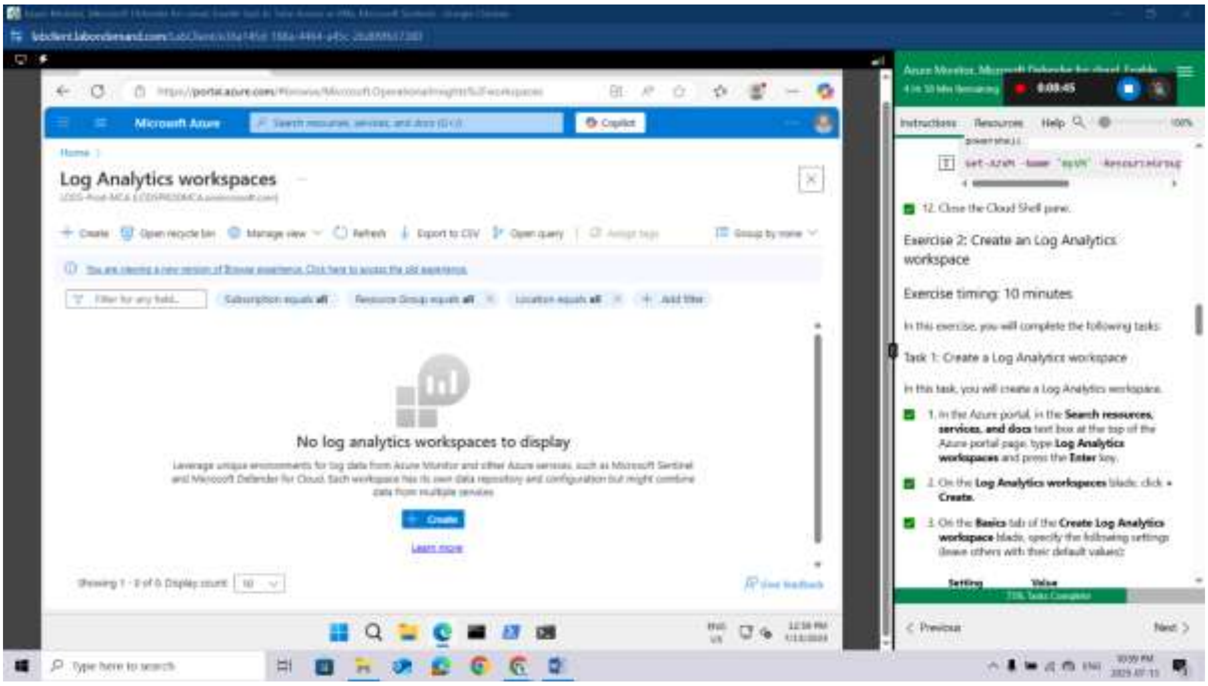
In this exercise, you will complete the following tasks:

Task 1: Create a Log Analytics workspace

In this task, you will create a Log Analytics workspace.

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Log Analytics workspaces** and press the **Enter** key.

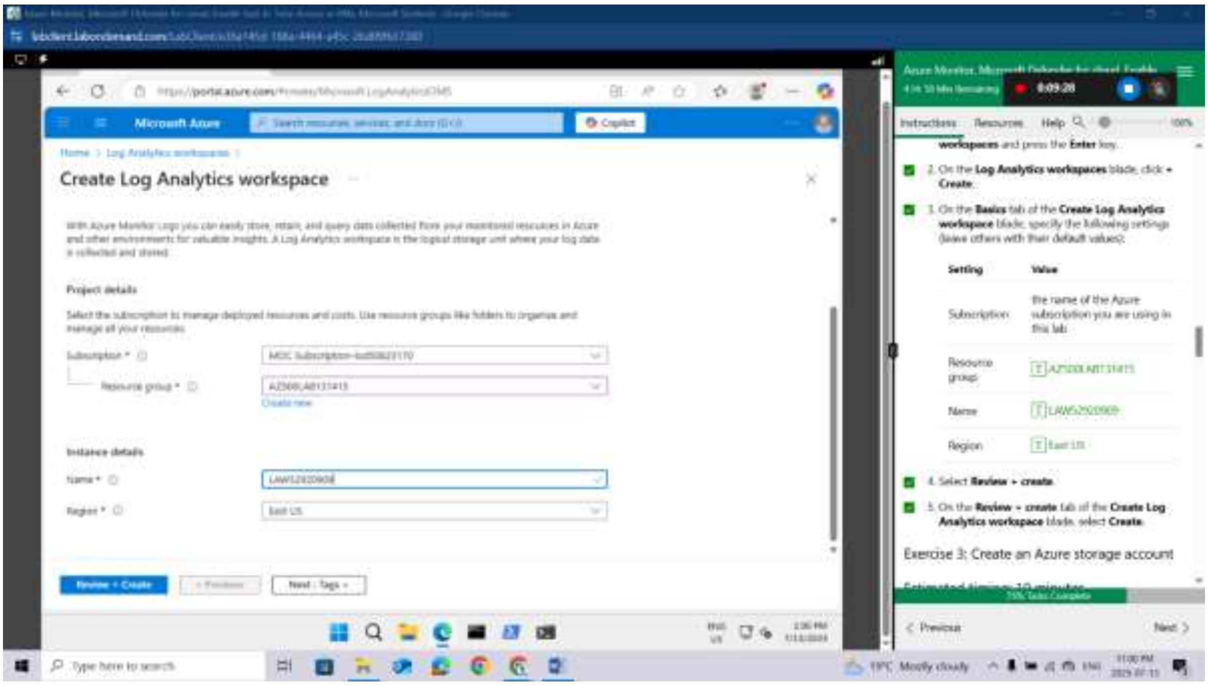
On the **Log Analytics workspaces** blade, click **+ Create**.



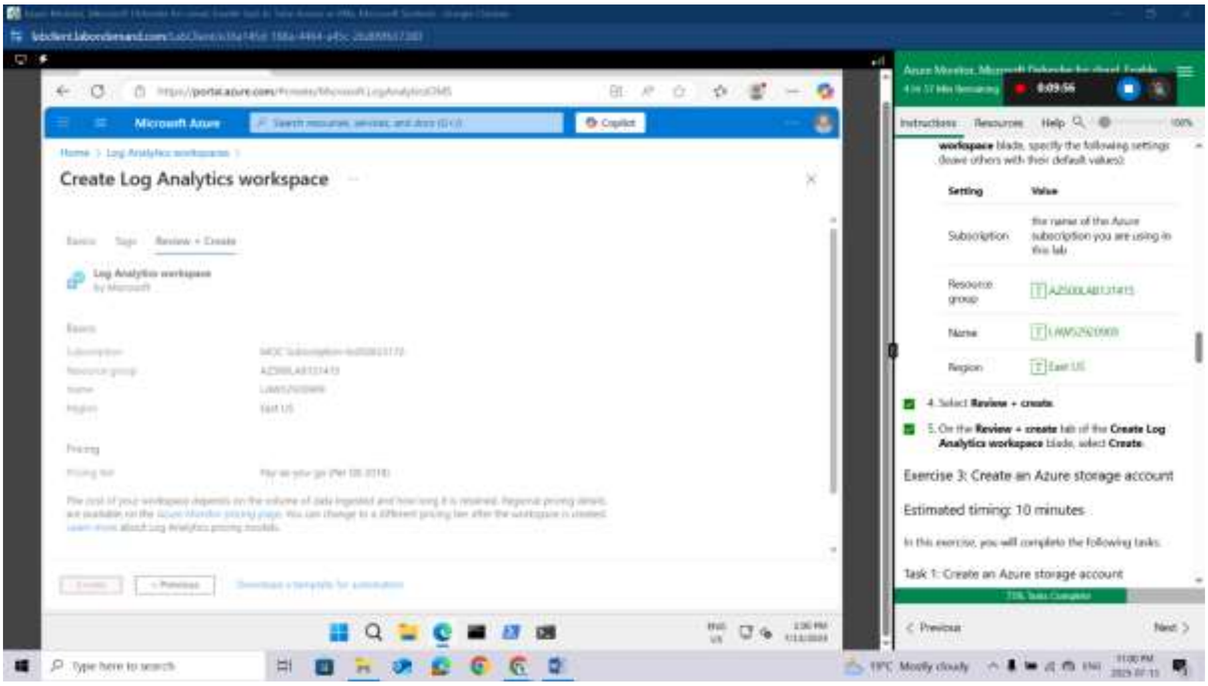
On the **Basics** tab of the **Create Log Analytics workspace** blade, specify the following settings (leave others with their default values):

Setting	Value
Subscription	the name of the Azure subscription you are using in this lab
Resource group	AZ500LAB131415
Name	LAW52920909
Region	East US

Select **Review + create**.



On the **Review + create** tab of the **Create Log Analytics workspace** blade, select **Create**.



Exercise 3: Create an Azure storage account

Estimated timing: 10 minutes

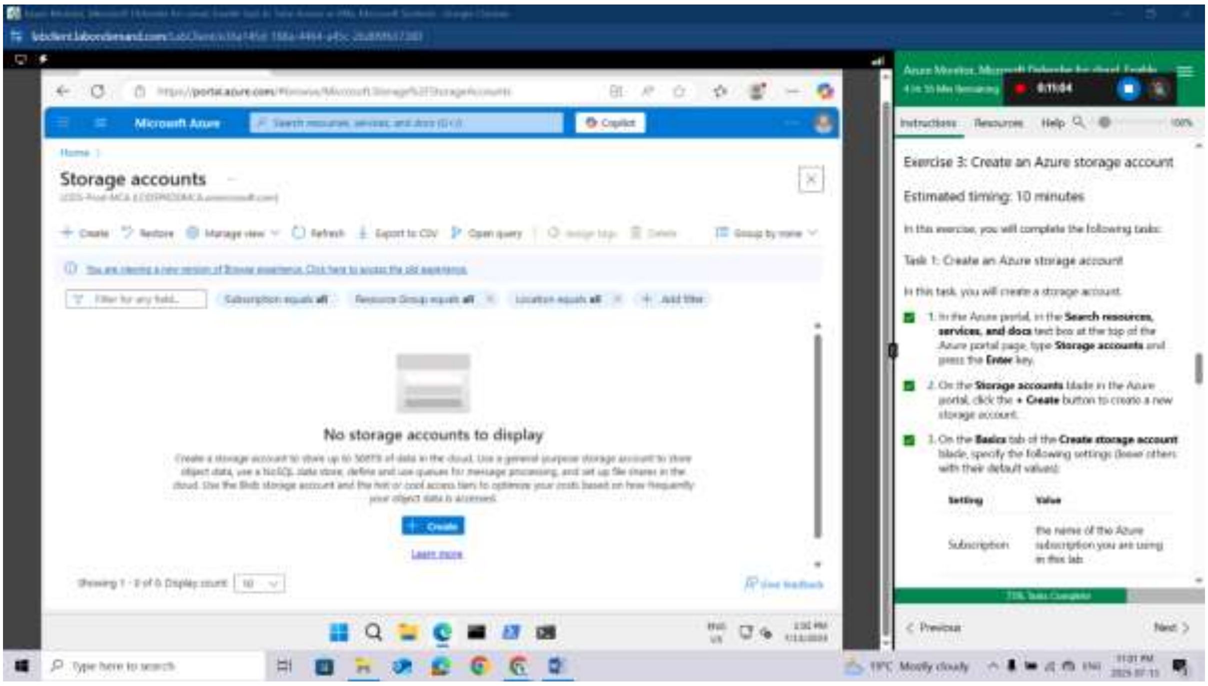
In this exercise, you will complete the following tasks:

Task 1: Create an Azure storage account

In this task, you will create a storage account.

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Storage accounts** and press the **Enter** key.

On the **Storage accounts** blade in the Azure portal, click the **+ Create** button to create a new storage account.

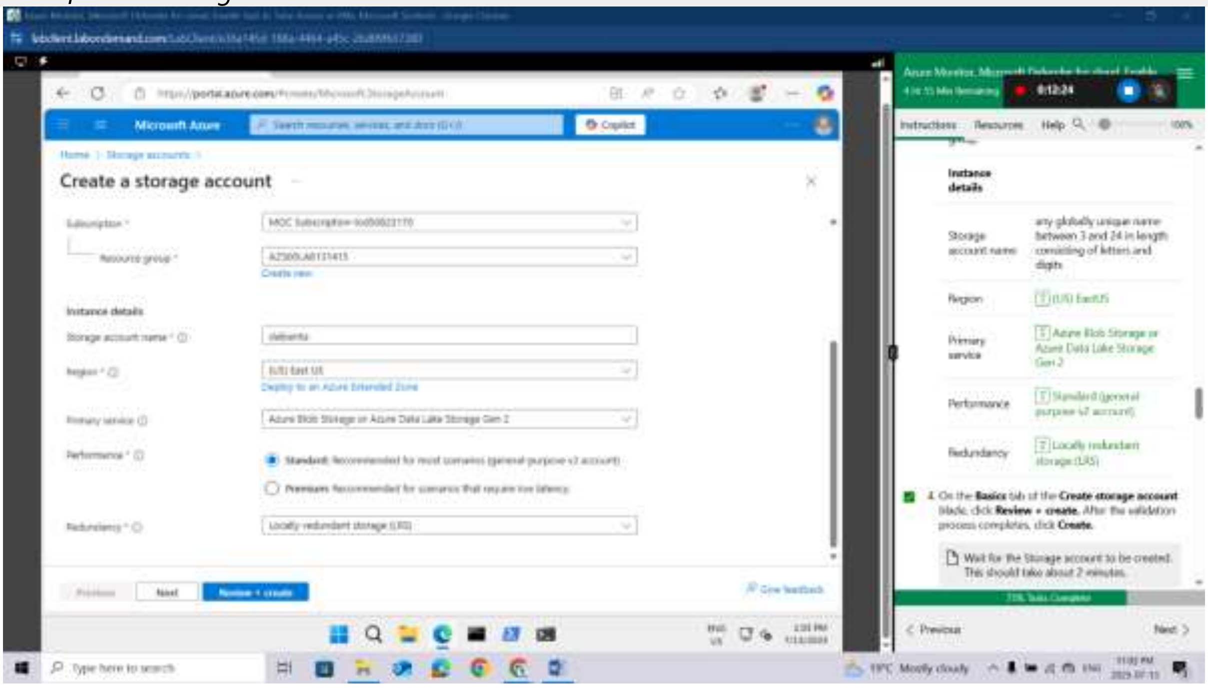


On the **Basics** tab of the **Create storage account** blade, specify the following settings (leave others with their default values):

Setting	Value
Subscription	the name of the Azure subscription you are using in this lab
Resource group	AZ500LAB131415
Instance details	
Storage account name	any globally unique name between 3 and 24 in length consisting of letters and digits
Region	(US) EastUS
Primary service	Azure Blob Storage or Azure Data Lake Storage Gen 2
Performance	Standard (general-purpose v2 account)
Redundancy	Locally redundant storage (LRS)

On the **Basics** tab of the **Create storage account** blade, click **Review + create.** After the validation process completes, click **Create.**

Wait for the Storage account to be created. This should take about 2 minutes.



Exercise 4: Create a Data Collection Rule

Estimated timing: 15 minutes

In this exercise, you will complete the following tasks:

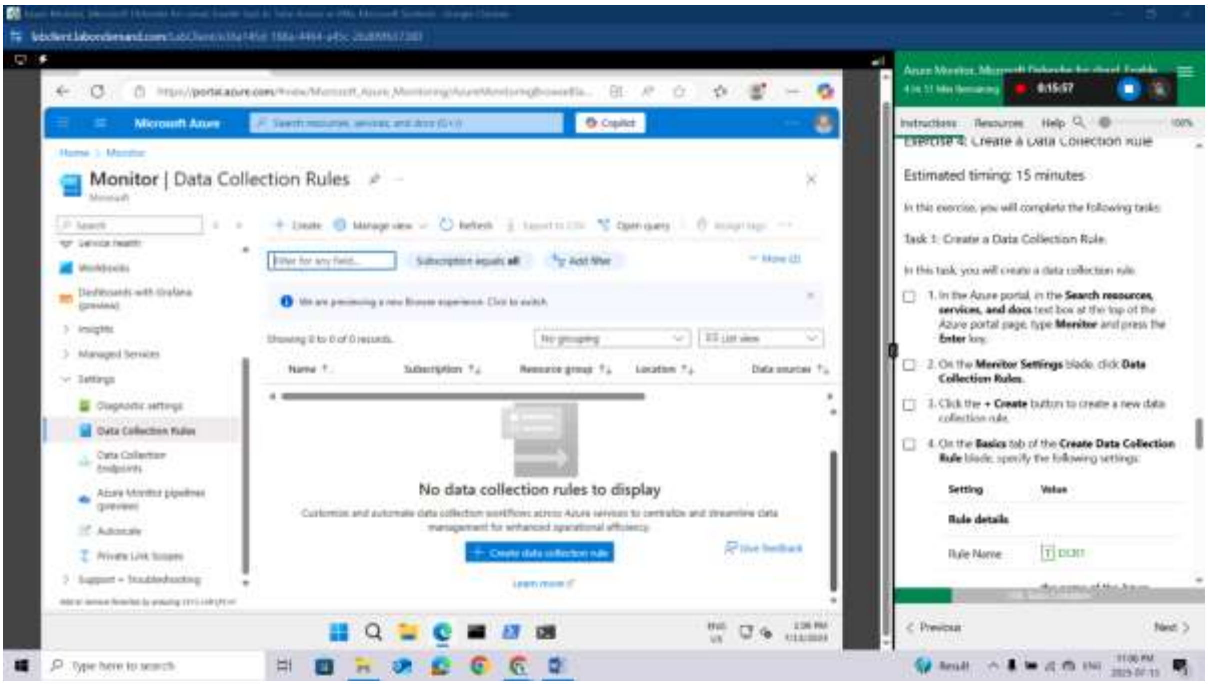
Task 1: Create a Data Collection Rule.

In this task, you will create a data collection rule.

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Monitor** and press the **Enter** key.

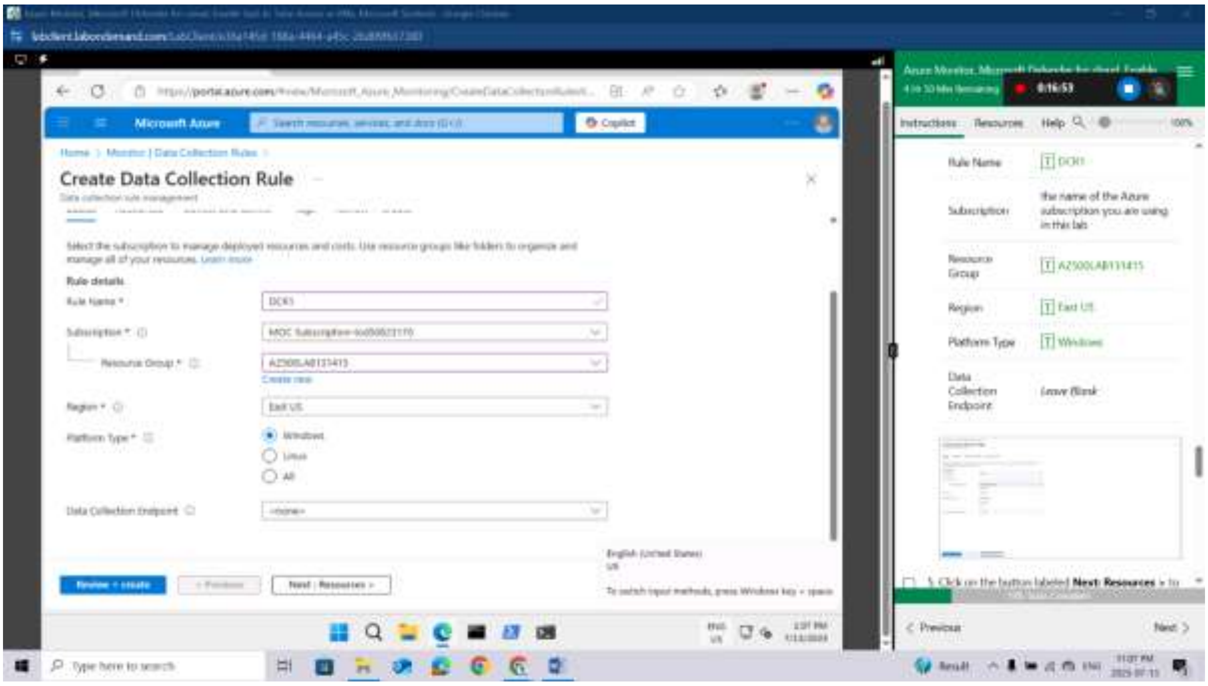
On the **Monitor Settings** blade, click **Data Collection Rules.**

Click the **+ Create** button to create a new data collection rule.



On the **Basics** tab of the **Create Data Collection Rule** blade, specify the following settings:

Setting	Value
Rule details	
Rule Name	DCR1
Subscription	the name of the Azure subscription you are using in this lab
Resource Group	AZ500LAB131415
Region	East US
Platform Type	Windows
Data Collection Endpoint	*Leave Blank*

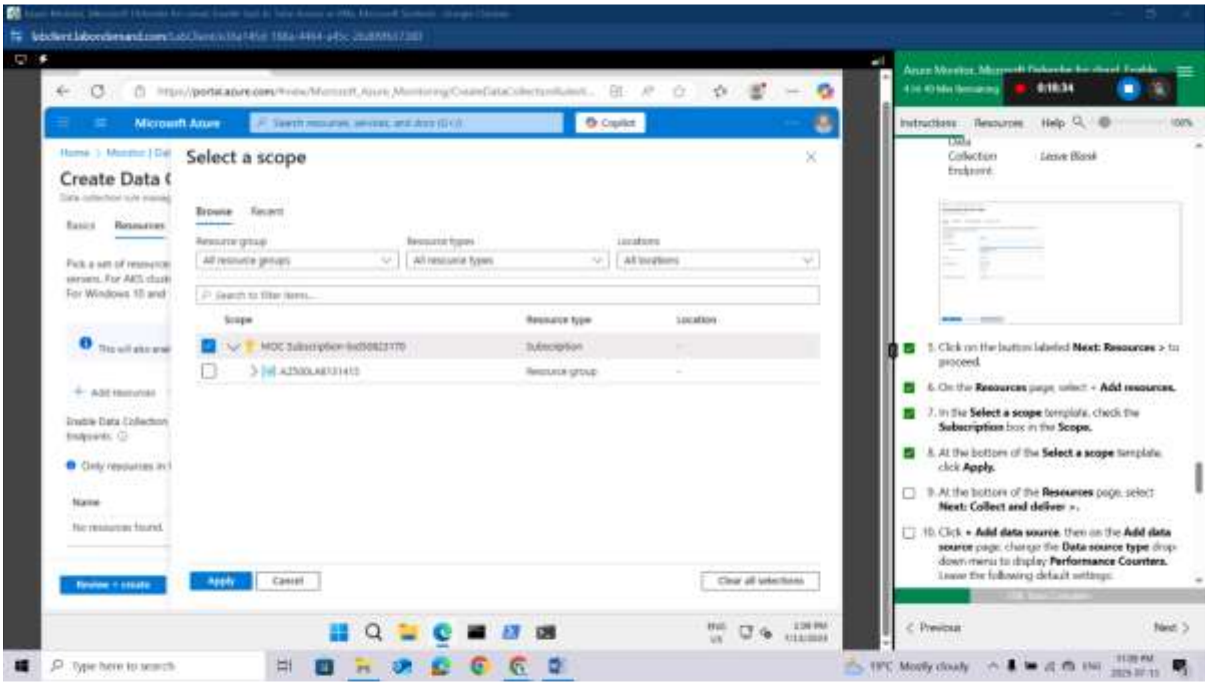


Click on the button labeled **Next: Resources >** to proceed.

On the **Resources** page, select + **Add resources.**

In the **Select a scope** template, check the **Subscription** box in the **Scope.**

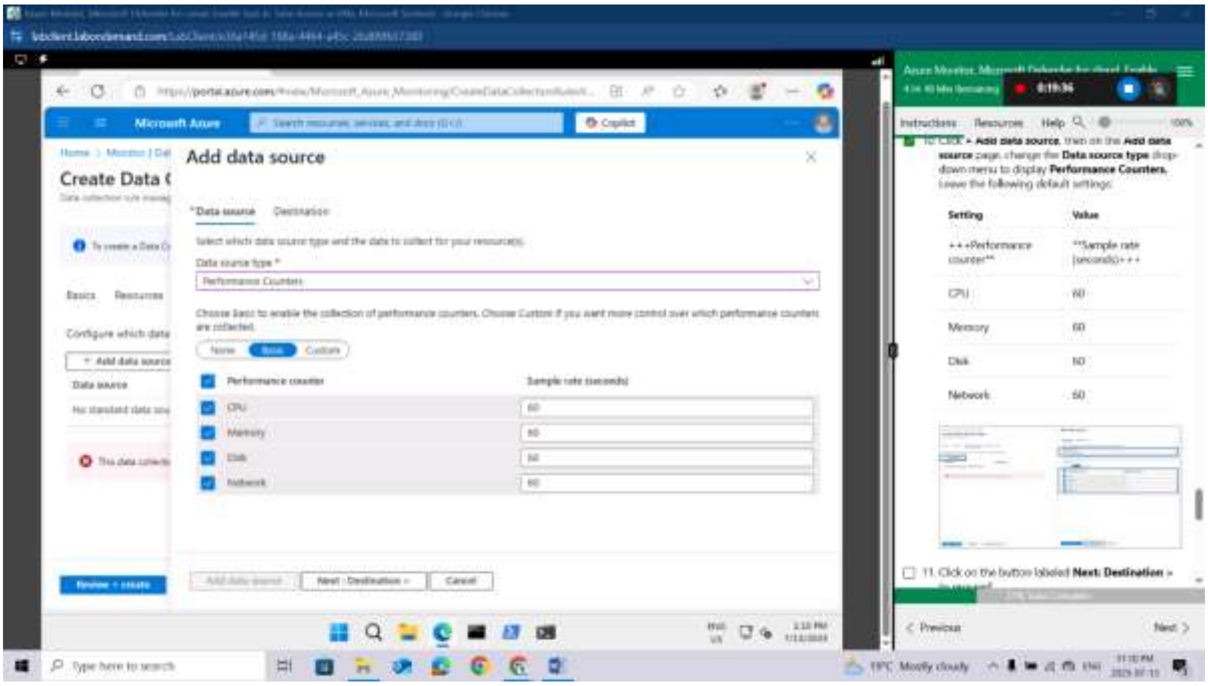
At the bottom of the **Select a scope** template, click **Apply.**



At the bottom of the **Resources** page, select **Next: Collect and deliver >.**

Click **+ Add data source**, then on the **Add data source** page, change the **Data source type** drop-down menu to display **Performance Counters.** Leave the following default settings:

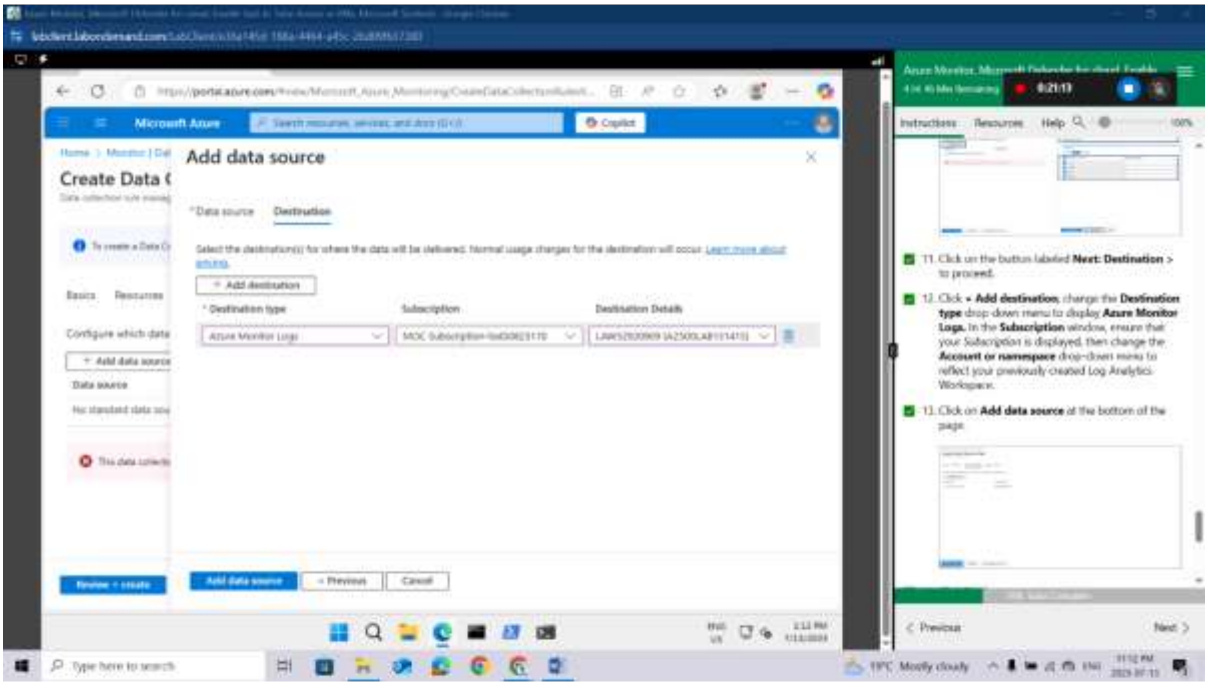
Setting	Value
+++Performance counter**	**Sample rate (seconds)+++
CPU	60
Memory	60
Disk	60
Network	60



Click on the button labeled **Next: Destination >** to proceed.

Click **+ Add destination**, change the **Destination type** drop-down menu to display **Azure Monitor Logs.** In the **Subscription** window, ensure that your *Subscription* is displayed, then change the **Account or namespace** drop-down menu to reflect your previously created Log Analytics Workspace.

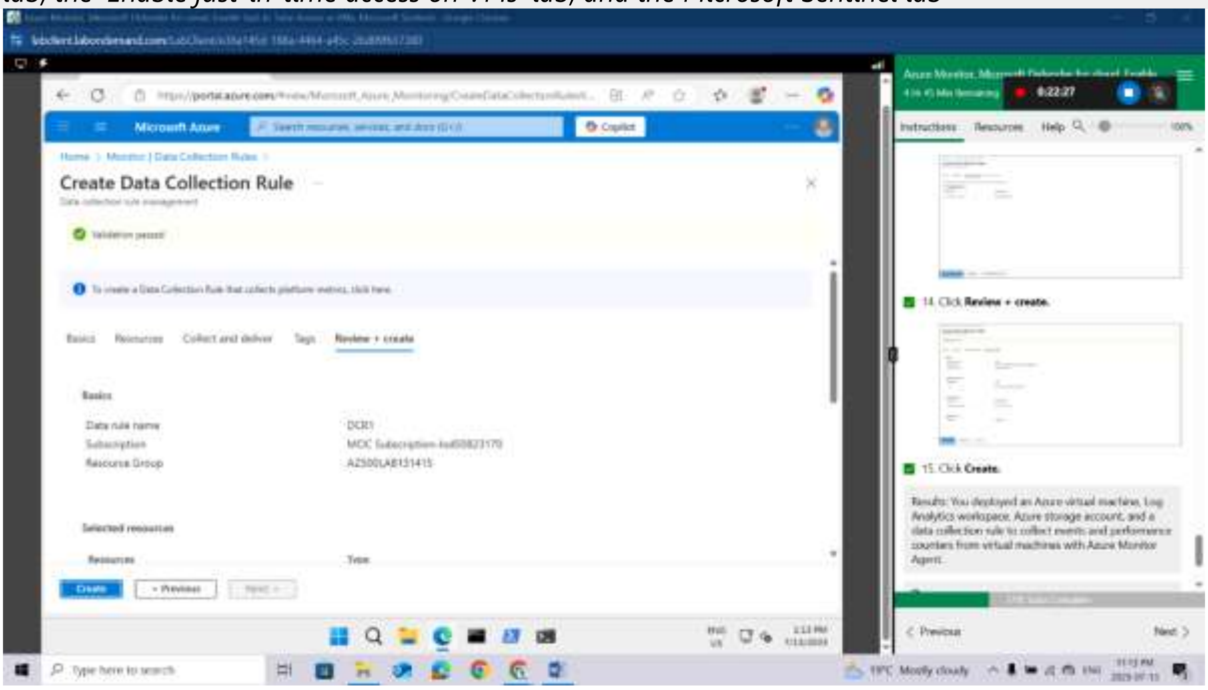
Click on **Add data source** at the bottom of the page.



Click **Review + create.**

Click **Create.**

Results: You deployed an Azure virtual machine, Log Analytics workspace, Azure storage account, and a data collection rule to collect events and performance counters from virtual machines with Azure Monitor Agent.
Do not remove the resources from this lab, as they are needed for the Microsoft Defender for Cloud lab, the 'Enable just-in-time access on VMs' lab, and the Microsoft Sentinel lab



Congratulations!You have successfully completed this Lab. Click **Next** to advance to the next **Lab**.

Lab 09: Configuring Microsoft Defender for Cloud Enhanced Security Features for Servers

Lab scenario

As an Azure Security Engineer for a global e-commerce company, you are responsible for securing the company's cloud infrastructure. The organization relies heavily on Azure virtual machines (VMs) and on-premises servers to run critical applications, manage customer data, and process transactions. The Chief Information Security Officer (CISO) has identified the need for enhanced security measures to protect these resources against cyber threats, vulnerabilities, and misconfigurations. You have been tasked with enabling Microsoft Defender for Servers in Microsoft Defender for Cloud to provide advanced threat protection and security monitoring for both Azure VMs and hybrid servers.

Lab objectives

- Configure Microsoft Defender for Cloud Enhanced Security Features for Servers
- Review the ehanced security features for Microsoft Defender for Servers Plan 2

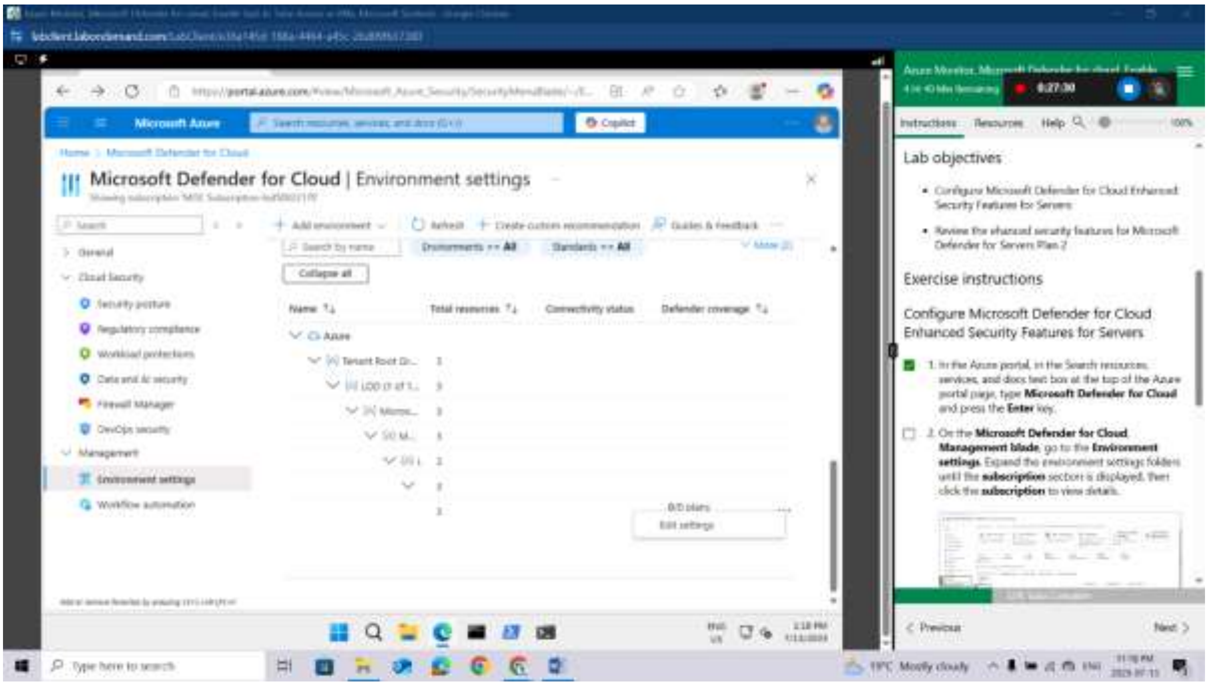
Exercise instructions

Configure Microsoft Defender for Cloud Enhanced Security Features for Servers

In the Azure portal, in the Search resources, services, and docs text box at the top of the Azure portal page, type **Microsoft Defender for Cloud** and press the **Enter** key.

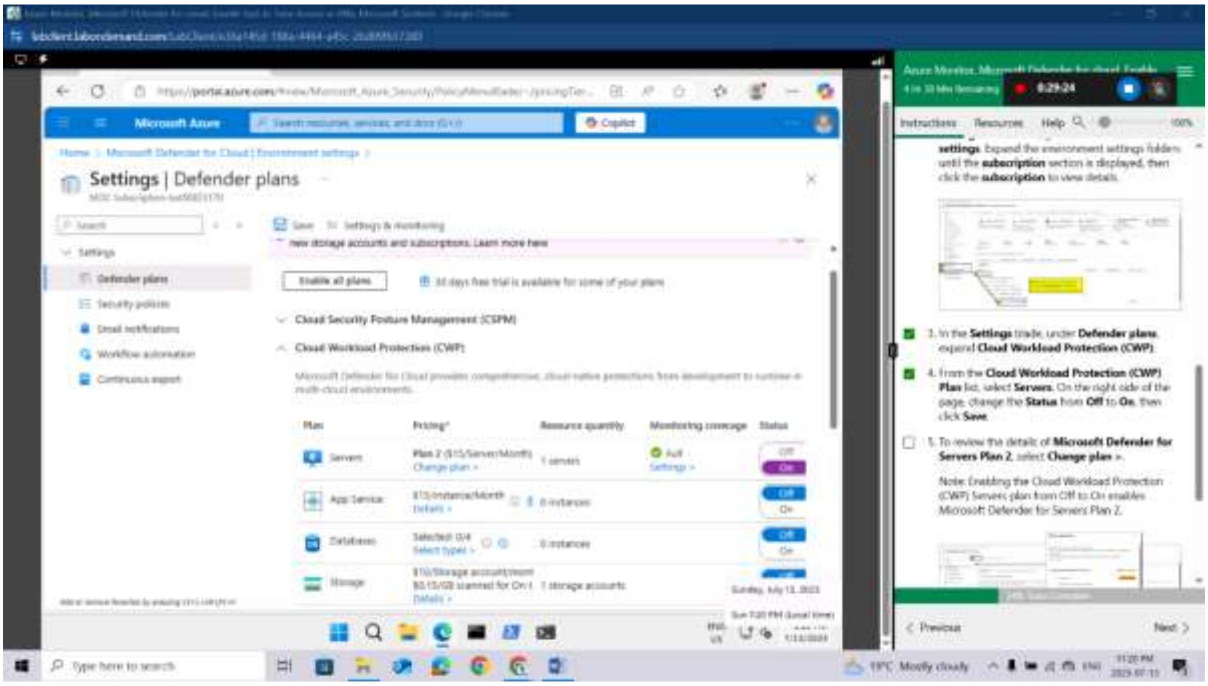
On the **Microsoft Defender for Cloud**, **Management blade**, go to the **Environment settings**. Expand the environment settings folders until the **subscription** section is displayed, then click the **subscription** to view details.

Click on editsettings to navigate through the subscription



In the **Settings** blade, under **Defender plans**, expand **Cloud Workload Protection (CWP)**.

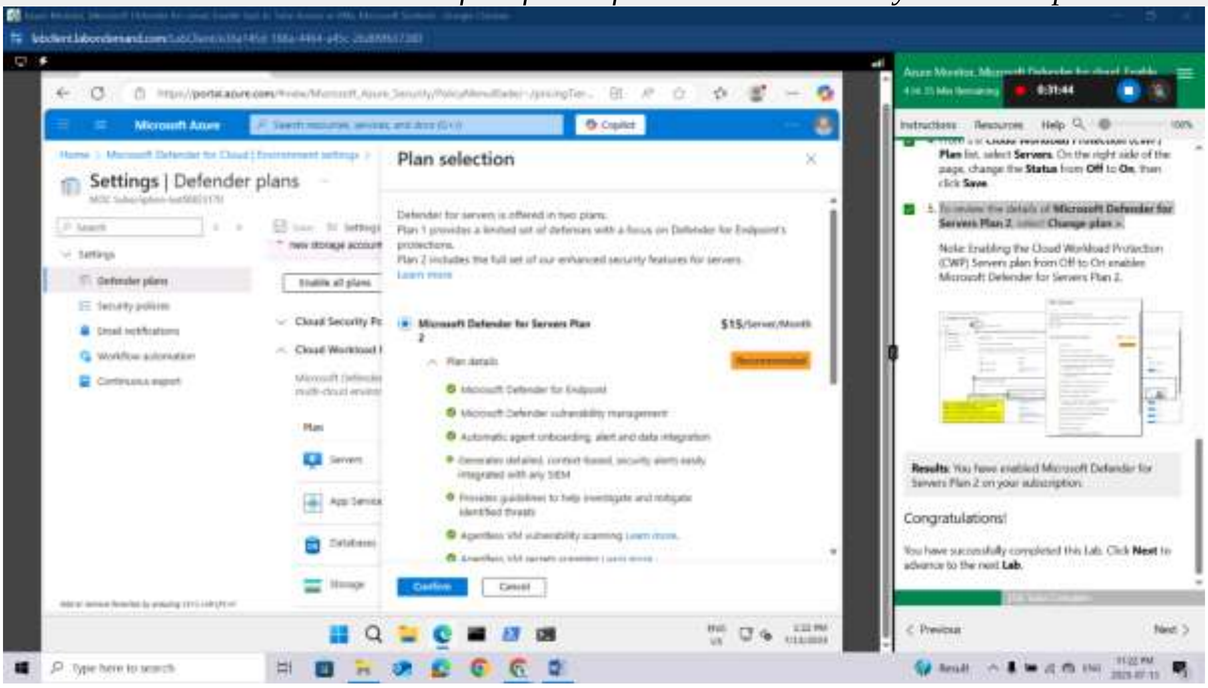
From the **Cloud Workload Protection (CWP) Plan** list, select **Servers**. On the right side of the page, change the **Status** from **Off** to **On**, then click **Save**.



To review the details of **Microsoft Defender for Servers Plan 2**, select **Change plan >**.

Note: Enabling the Cloud Workload Protection (CWP) Servers plan from Off to On enables Microsoft Defender for Servers Plan 2.

Results: *You have enabled Microsoft Defender for Servers Plan 2 on your subscription.*



Congratulations!You have successfully completed this Lab. Click **Next** to advance to the next **Lab**.

Lab 10: Enable just-in-time access on VMs

Lab scenario

As an Azure Security Engineer at a financial services company, you're responsible for securing Azure resources, including virtual machines (VMs) that host critical applications. The security team has identified that continuous open access to VMs increases the risk of brute-force attacks and unauthorized access. To mitigate this, the Chief Information Security Officer (CISO) has requested that you enable Just-in-Time (JIT) VM access on a specific Azure VM used for processing financial transactions.

Lab objectives

In this lab, you will complete the following exercises:

- Exercise 1: Enable JIT on your VMs from the Azure portal.
- Exercise 2: Request access to a VM that has JIT enabled from the Azure portal.

Exercise instructions

Exercise 1: Enable JIT on your VMs from Azure virtual machines

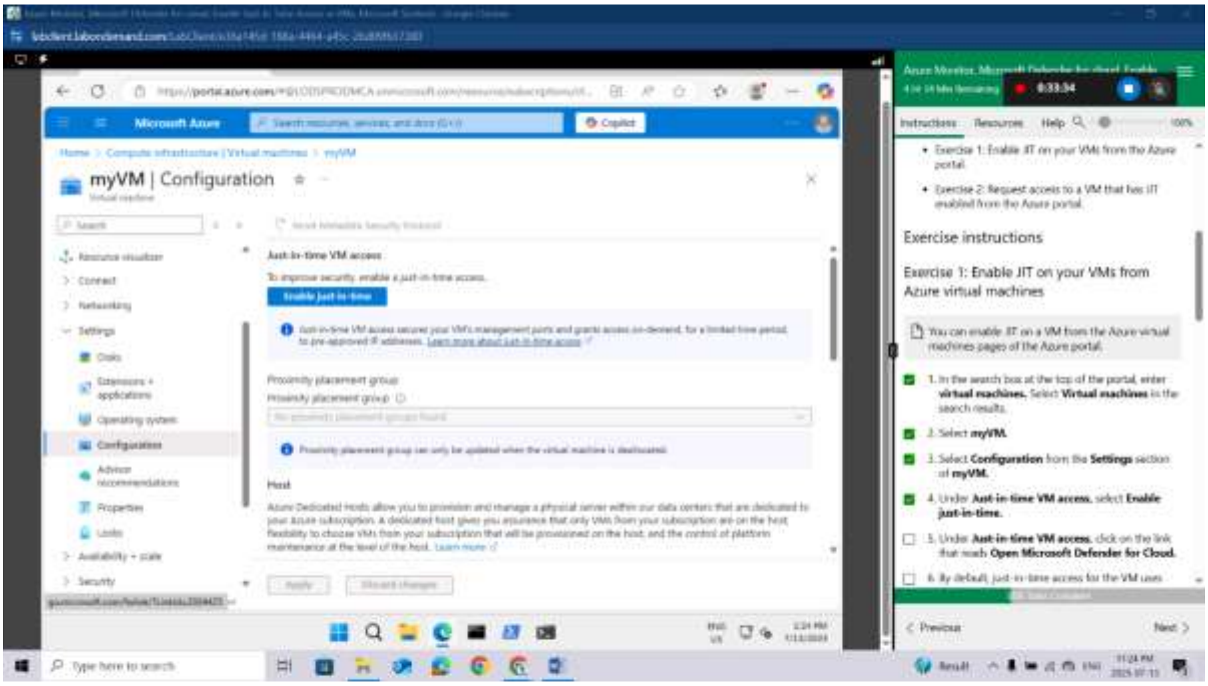
You can enable JIT on a VM from the Azure virtual machines pages of the Azure portal.

In the search box at the top of the portal, enter **virtual machines.** Select **Virtual machines** in the search results.

Select **myVM.**

Select **Configuration** from the **Settings** section of **myVM.**

Under **Just-in-time VM access,** select **Enable just-in-time.**



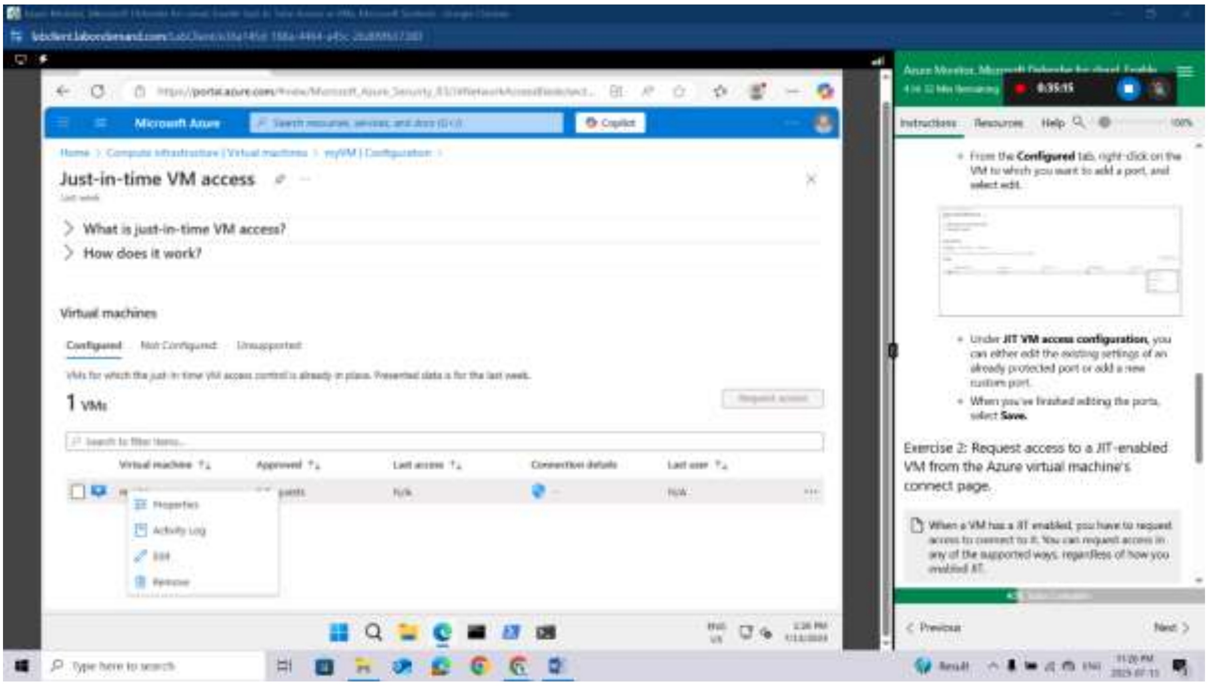
Under **Just-in-time VM access,** click on the link that reads **Open Microsoft Defender for Cloud.**

By default, just-in-time access for the VM uses these settings:

- Windows machines
  - RDP port: 3389
  - Maximum allowed access: Three hours
  - Allowed source IP addresses: Any
- Linux machines
  - SSH port: 22
  - Maximum allowed access: Three hours
  - Allowed source IP addresses: Any

By default, just-in-time access for the VM uses these settings:

- From the **Configured** tab, right-click on the VM to which you want to add a port, and select edit.



- Under **JIT VM access configuration,** you can either edit the existing settings of an already protected port or add a new custom port.
- When you've finished editing the ports, select **Save.**

Exercise 2: Request access to a JIT-enabled VM from the Azure virtual machine's connect page.

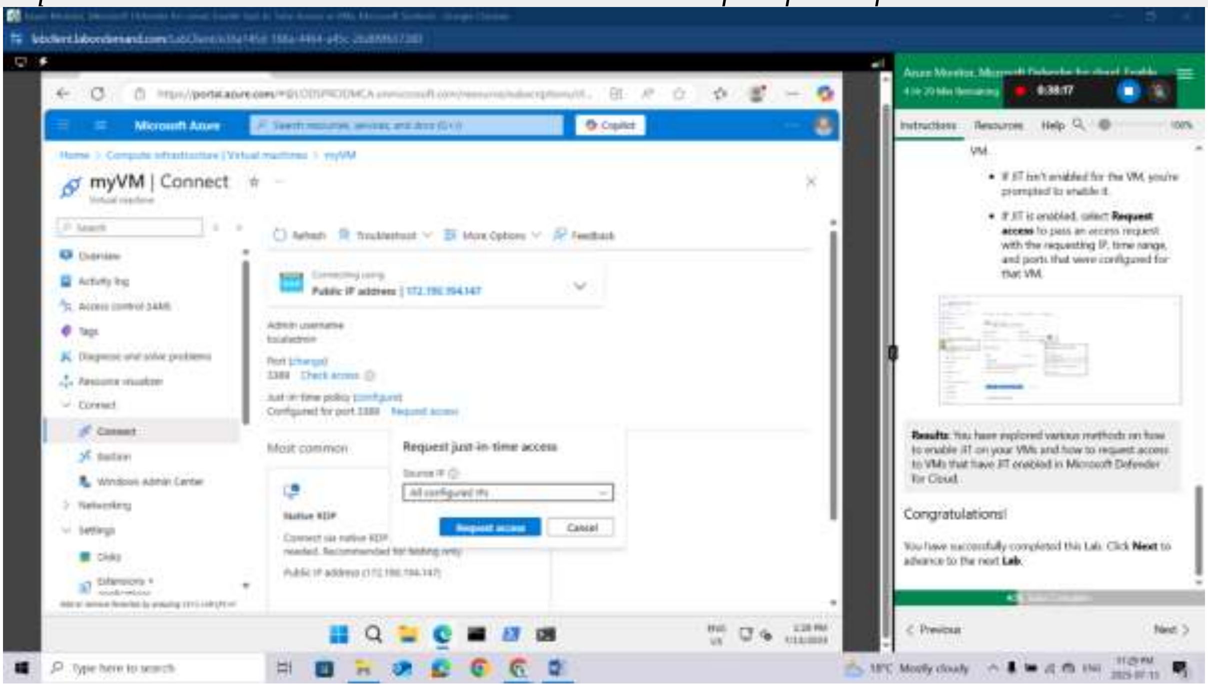
When a VM has a JIT enabled, you have to request access to connect to it. You can request access in any of the supported ways, regardless of how you enabled JIT.

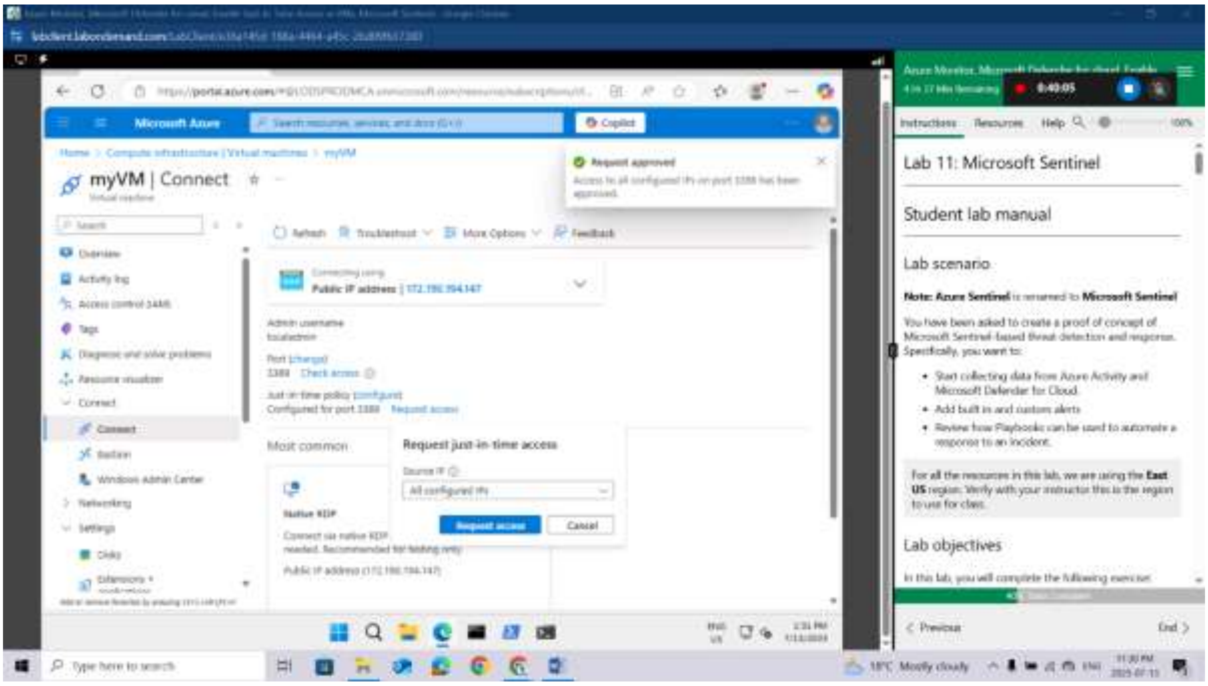
In the Azure portal, open the virtual machines pages.

Select the VM to which you want to connect, and open the **Connect** page.

- Azure checks to see if JIT is enabled on that VM.
  - If JIT isn't enabled for the VM, you're prompted to enable it.
  - If JIT is enabled, select **Request access** to pass an access request with the requesting IP, time range, and ports that were configured for that VM.

Results*: You have explored various methods on how to enable JIT on your VMs and how to request access to VMs that have JIT enabled in Microsoft Defender for Cloud.*





Congratulations!You have successfully completed this Lab. Click **Next** to advance to the next **Lab**.

Lab 11: Microsoft Sentinel
Lab scenario

Note: Azure Sentinel is renamed to **Microsoft Sentinel**

You have been asked to create a proof of concept of Microsoft Sentinel-based threat detection and response. Specifically, you want to:

- Start collecting data from Azure Activity and Microsoft Defender for Cloud.
- Add built in and custom alerts
- Review how Playbooks can be used to automate a response to an incident.

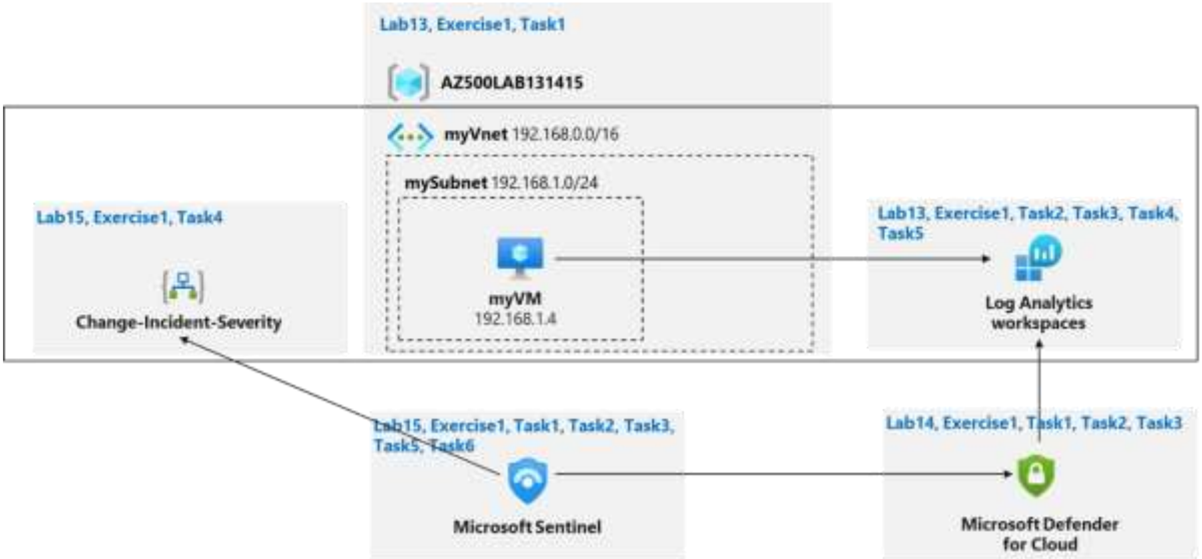
*For all the resources in this lab, we are using the **East US** region. Verify with your instructor this is the region to use for class.*

Lab objectives

In this lab, you will complete the following exercise:

- Exercise 1: Implement Microsoft Sentinel

Microsoft Sentinel diagram



Instructions

Lab files:

- **\Allfiles\Labs\15\changeincidentseverity.json**

Exercise 1: Implement Microsoft Sentinel

Estimated timing: 30 minutes

In this exercise, you will complete the following tasks:

- Task 1: On-board Microsoft Sentinel
- Task 2: Connect Azure Activity to Sentinel
- Task 3: Create a rule that uses the Azure Activity data connector.
- Task 4: Create a playbook
- Task 5: Create a custom alert and configure the playbook as an automated response.
- Task 6: Invoke an incident and review the associated actions.

Task 1: On-board Microsoft Sentinel

In this task, you will on-board Microsoft Sentinel and connect the Log Analytics workspace.

Sign-in to the Azure portal `https://portal.azure.com/`.

*Sign in to the Azure portal using an account that has the Owner or Contributor role in the Azure subscription you are using for this lab. In this **Cloudslice** lab, this account is LabUser-52920909@LODSPRODMCA.onmicrosoft.com with password ADg07!e$kHK@.*

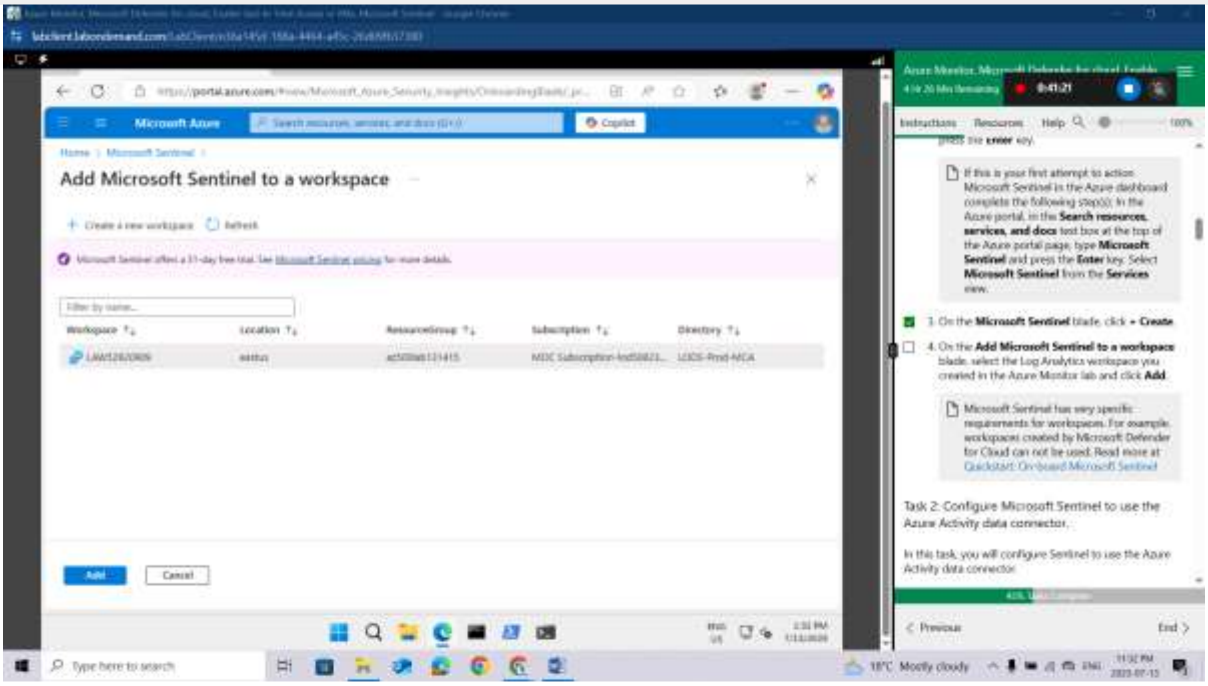
In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Microsoft Sentinel** and press the **Enter** key.

*If this is your first attempt to action Microsoft Sentinel in the Azure dashboard complete the following step(s): In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Microsoft Sentinel** and press the **Enter** key. Select **Microsoft Sentinel** from the **Services** view.*

On the **Microsoft Sentinel** blade, click **+ Create**.

On the **Add Microsoft Sentinel to a workspace** blade, select the Log Analytics workspace you created in the Azure Monitor lab and click **Add**.

Microsoft Sentinel has very specific requirements for workspaces. For example, workspaces created by Microsoft Defender for Cloud can not be used. Read more at [Quickstart: On-board Microsoft Sentinel](#)

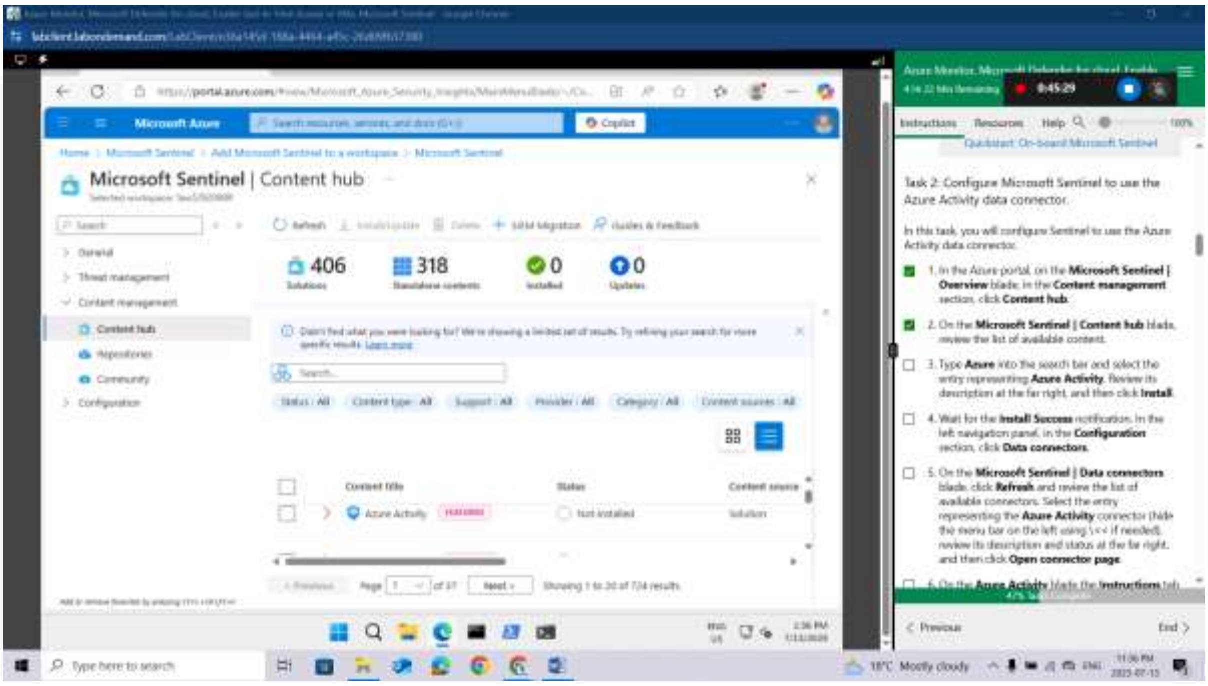


Task 2: Configure Microsoft Sentinel to use the Azure Activity data connector.

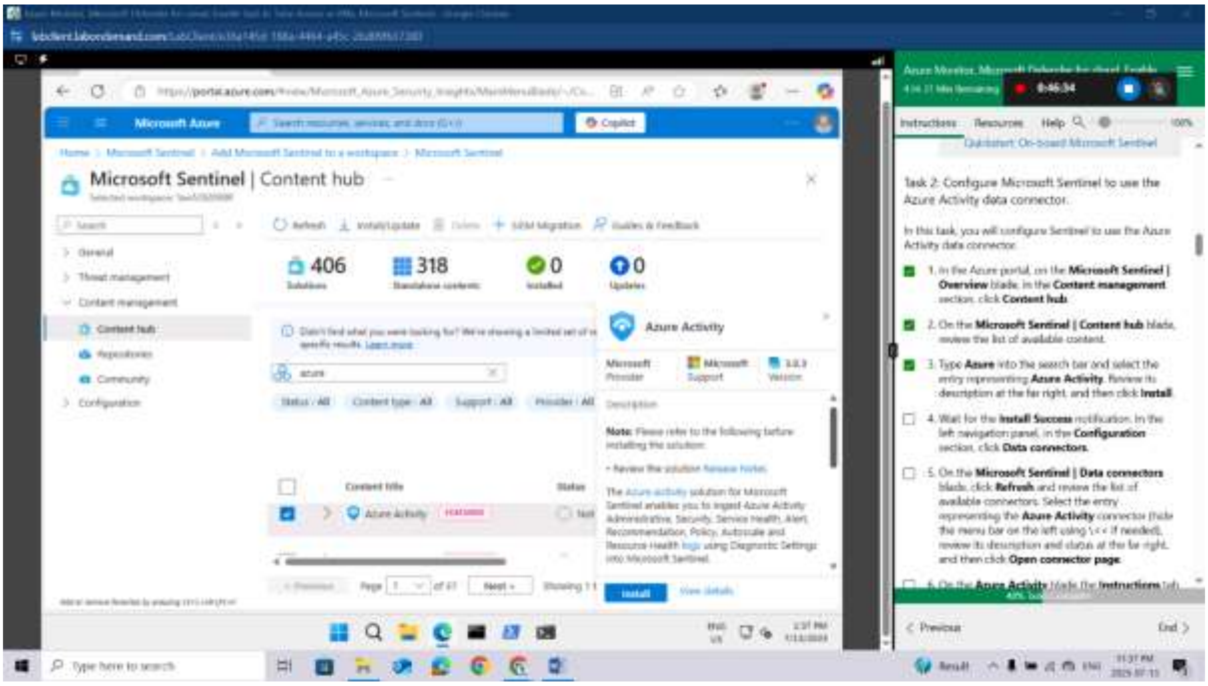
In this task, you will configure Sentinel to use the Azure Activity data connector.

In the Azure portal, on the **Microsoft Sentinel | Overview** blade, in the **Content management** section, click **Content hub**.

On the **Microsoft Sentinel | Content hub** blade, review the list of available content.

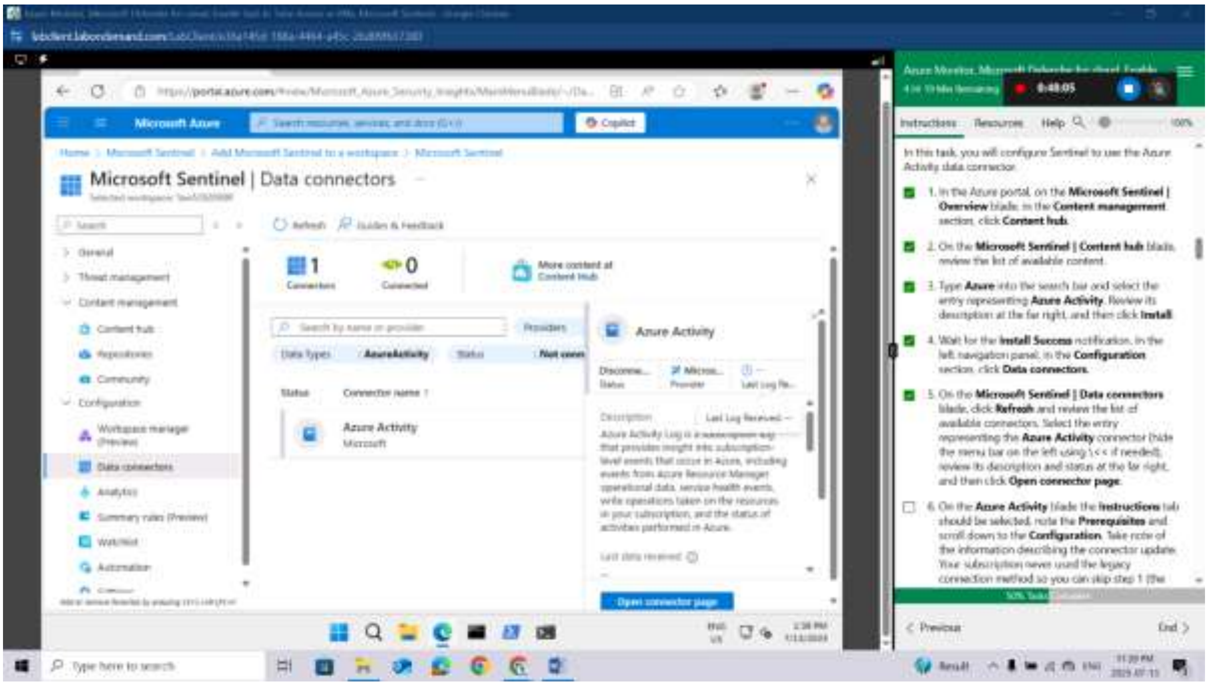


Type **Azure** into the search bar and select the entry representing **Azure Activity**. Review its description at the far right, and then click **Install**.



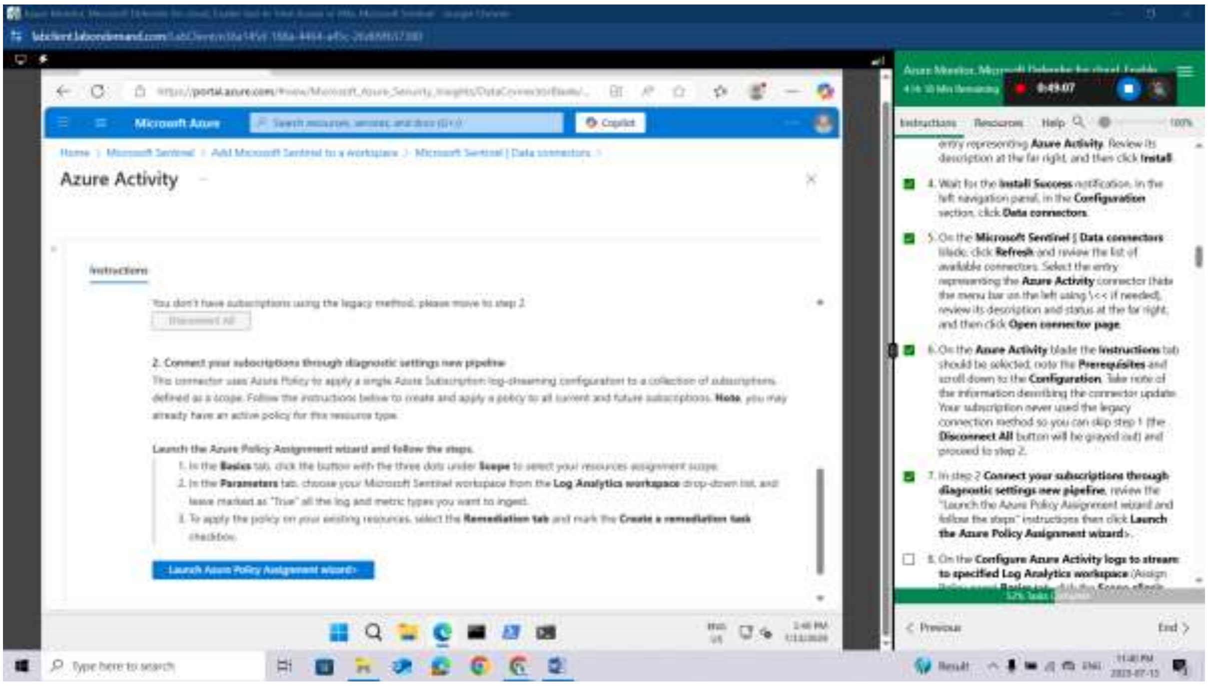
Wait for the **Install Success** notification. In the left navigation panel, in the **Configuration** section, click **Data connectors**.

On the **Microsoft Sentinel | Data connectors** blade, click **Refresh** and review the list of available connectors. Select the entry representing the **Azure Activity** connector (hide the menu bar on the left using \<< if needed), review its description and status at the far right, and then click **Open connector page**.



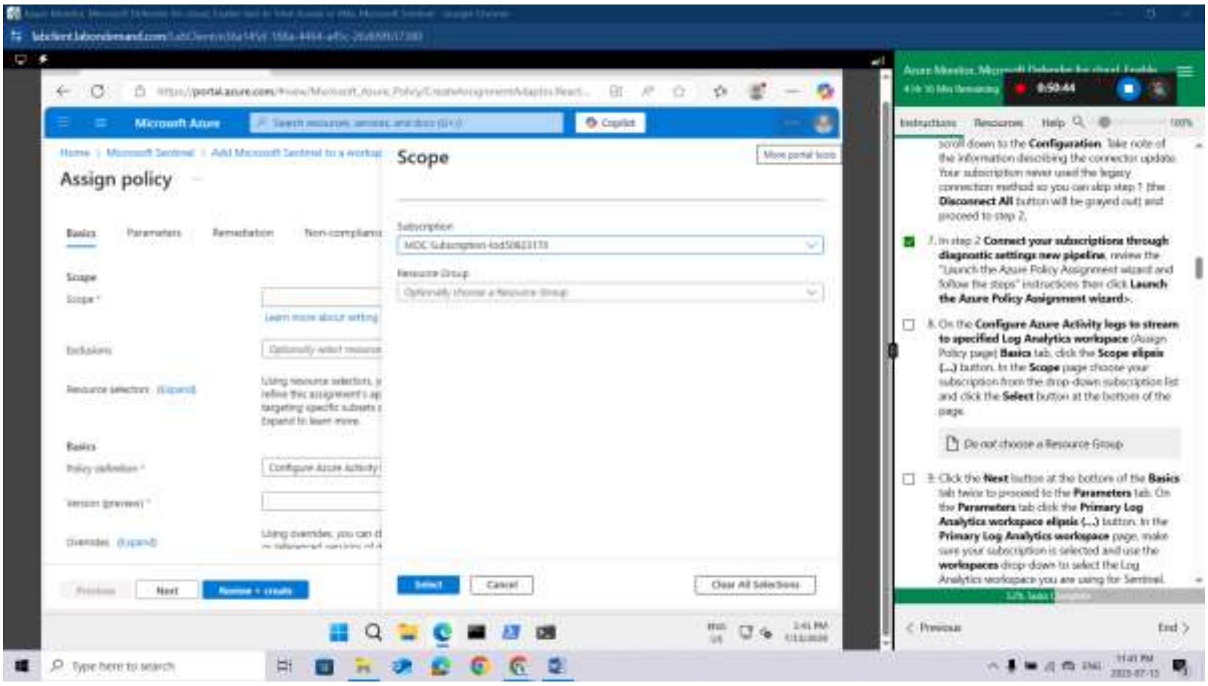
On the **Azure Activity** blade the **Instructions** tab should be selected, note the **Prerequisites** and scroll down to the **Configuration**. Take note of the information describing the connector update. Your subscription never used the legacy connection method so you can skip step 1 (the **Disconnect All** button will be grayed out) and proceed to step 2.

In step 2 **Connect your subscriptions through diagnostic settings new pipeline**, review the "Launch the Azure Policy Assignment wizard and follow the steps" instructions then click **Launch the Azure Policy Assignment wizard>**.

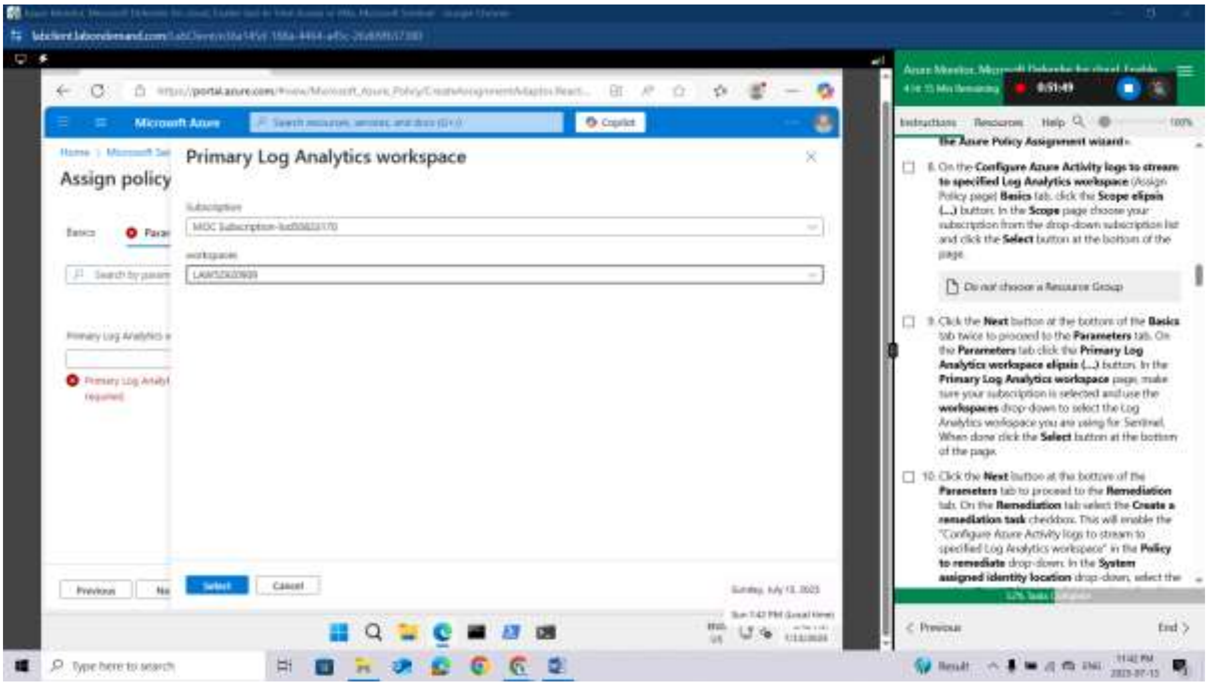


On the **Configure Azure Activity logs to stream to specified Log Analytics workspace** (Assign Policy page) **Basics** tab, click the **Scope elipsis (...)** button. In the **Scope** page choose your subscription from the drop-down subscription list and click the **Select** button at the bottom of the page.

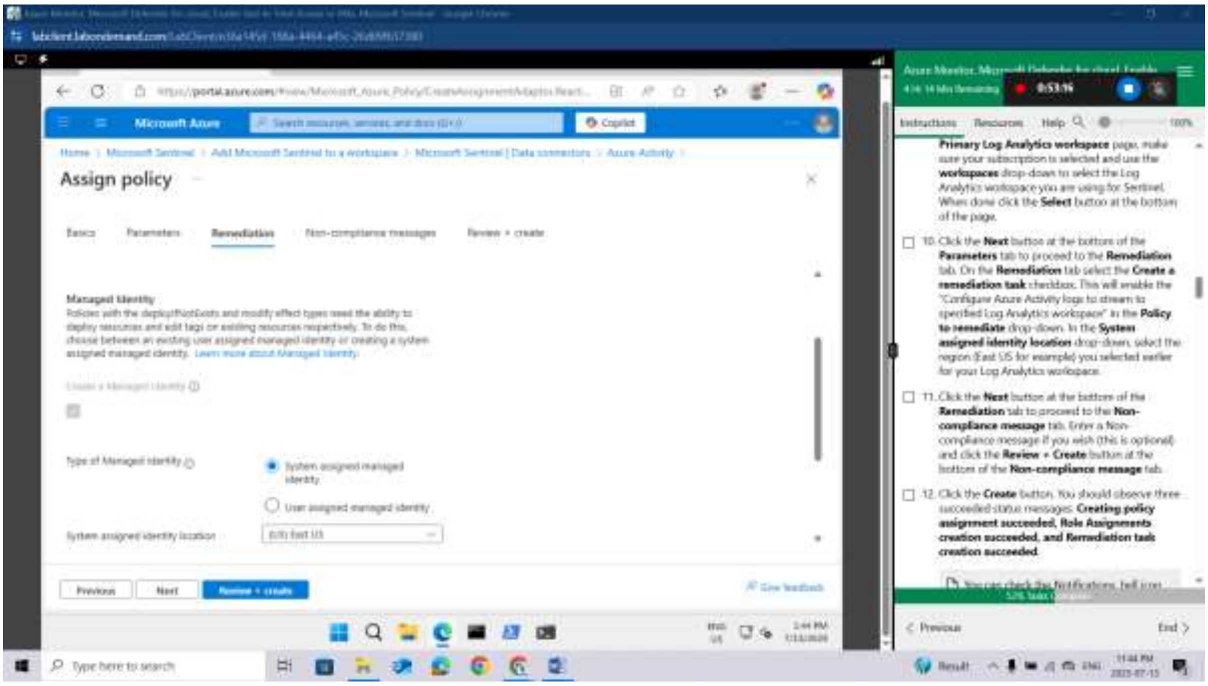
Do not choose a Resource Group



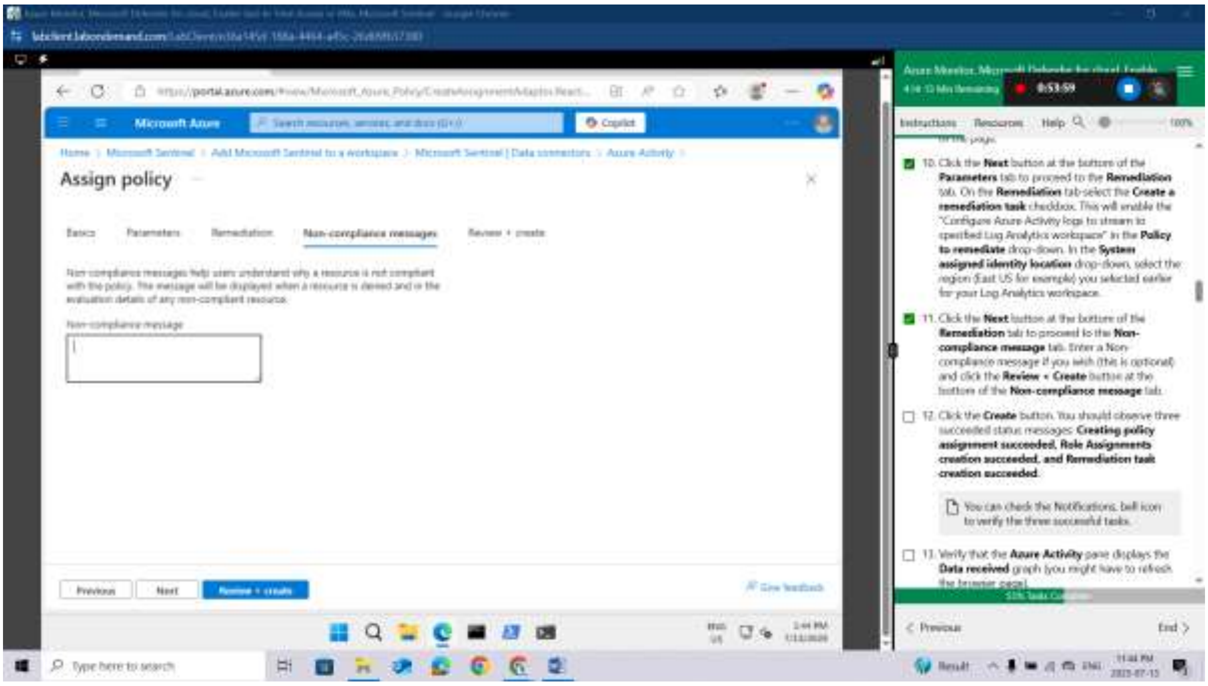
Click the **Next** button at the bottom of the **Basics** tab twice to proceed to the **Parameters** tab. On the **Parameters** tab click the **Primary Log Analytics workspace elipsis (...)** button. In the **Primary Log Analytics workspace** page, make sure your subscription is selected and use the **workspaces** drop-down to select the Log Analytics workspace you are using for Sentinel. When done click the **Select** button at the bottom of the page.



Click the **Next** button at the bottom of the **Parameters** tab to proceed to the **Remediation** tab. On the **Remediation** tab select the **Create a remediation task** checkbox. This will enable the "Configure Azure Activity logs to stream to specified Log Analytics workspace" in the **Policy to remediate** drop-down. In the **System assigned identity location** drop-down, select the region (East US for example) you selected earlier for your Log Analytics workspace.

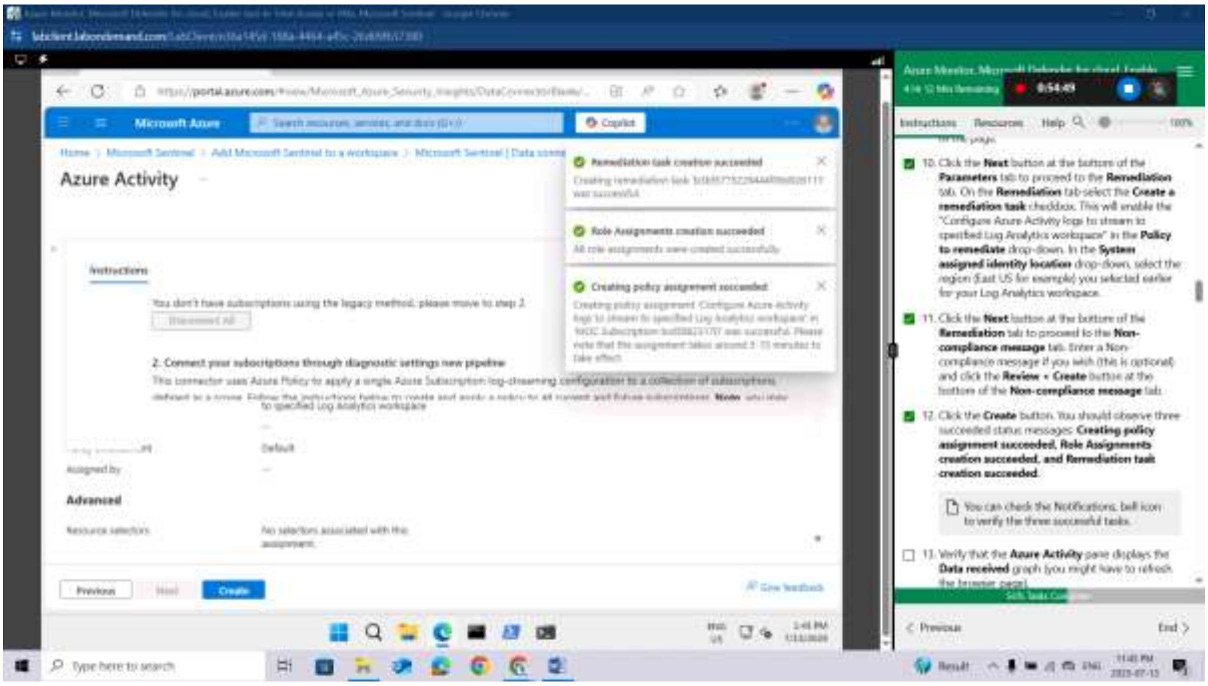


Click the **Next** button at the bottom of the **Remediation** tab to proceed to the **Non-compliance message** tab. Enter a Non-compliance message if you wish (this is optional) and click the **Review + Create** button at the bottom of the **Non-compliance message** tab.



Click the **Create** button. You should observe three succeeded status messages: **Creating policy assignment succeeded, Role Assignments creation succeeded, and Remediation task creation succeeded**.

You can check the Notifications, bell icon to verify the three successful tasks.



Verify that the **Azure Activity** pane displays the **Data received** graph (you might have to refresh the browser page).

It may take over 15 minutes before the Status shows "Connected" and the graph displays Data received.

Task 3: Create a rule that uses the Azure Activity data connector.

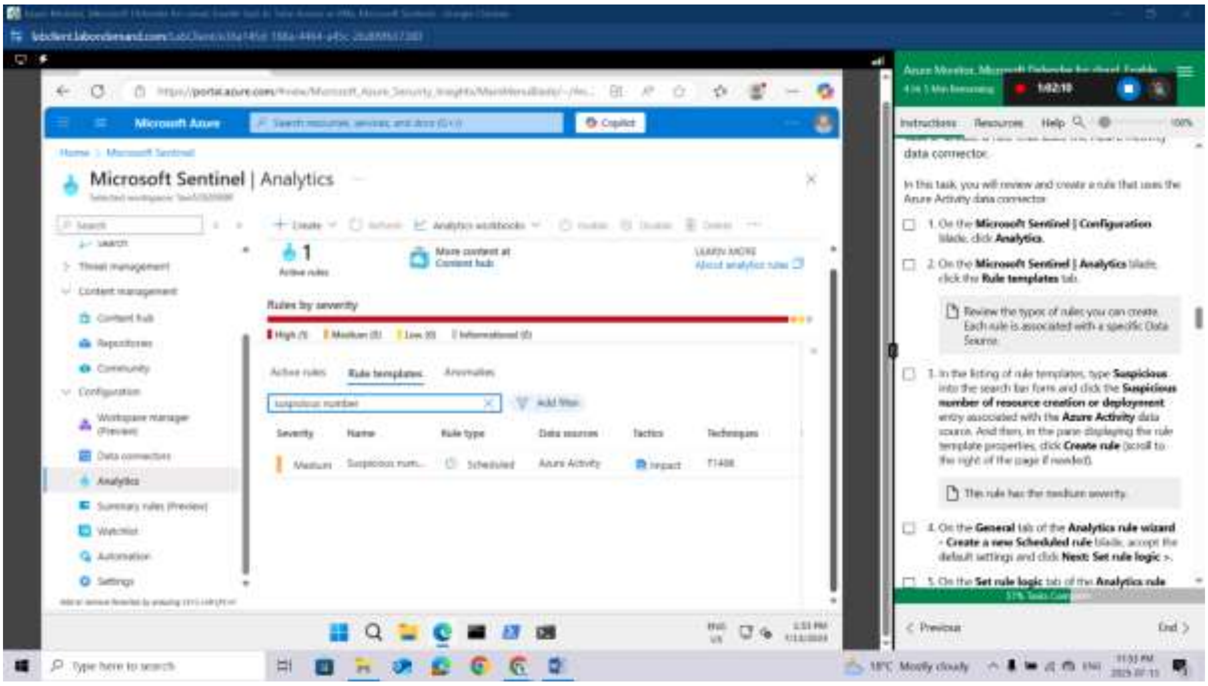
In this task, you will review and create a rule that uses the Azure Activity data connector.

On the **Microsoft Sentinel | Configuration** blade, click **Analytics**.

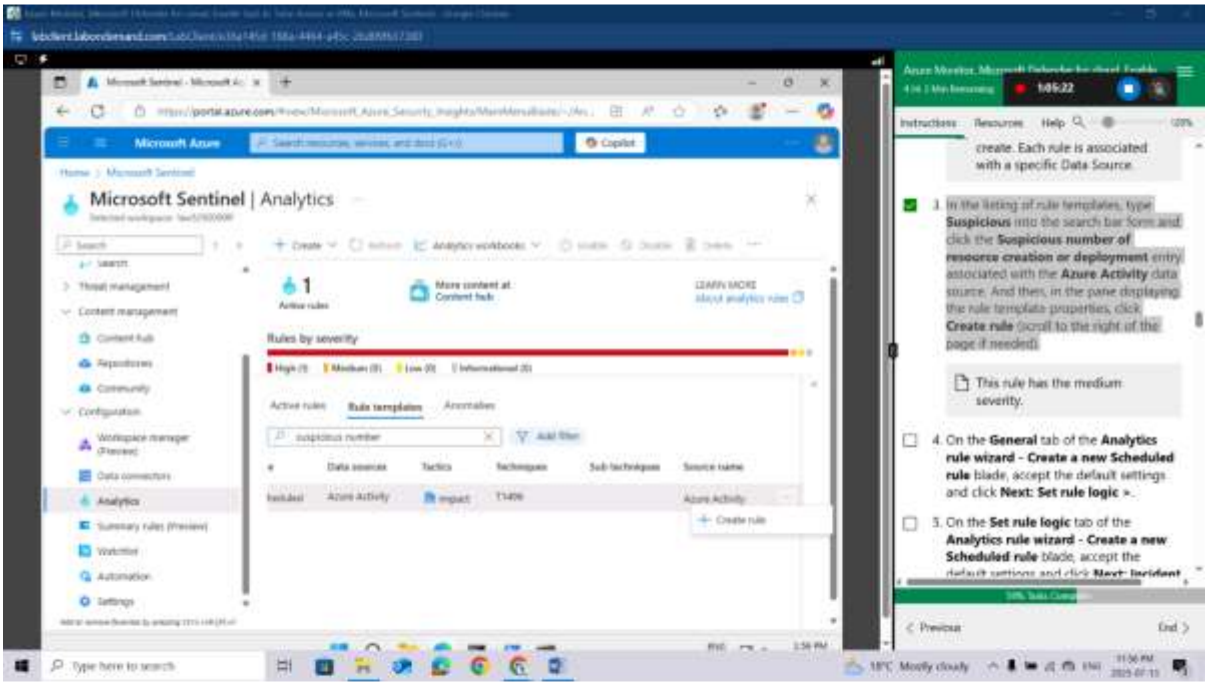
On the **Microsoft Sentinel | Analytics** blade, click the **Rule templates** tab.

Review the types of rules you can create. Each rule is associated with a specific Data Source.

In the listing of rule templates, type **Suspicious** into the search bar form and click the **Suspicious number of resource creation or deployment** entry associated with the **Azure Activity** data source. And then, in the pane displaying the rule template properties, click **Create rule** (scroll to the right of the page if needed).



This rule has the medium severity.



On the **General** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, accept the default settings and click **Next: Set rule logic >**.

On the **Set rule logic** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, accept the default settings and click **Next: Incident settings (Preview) >**.

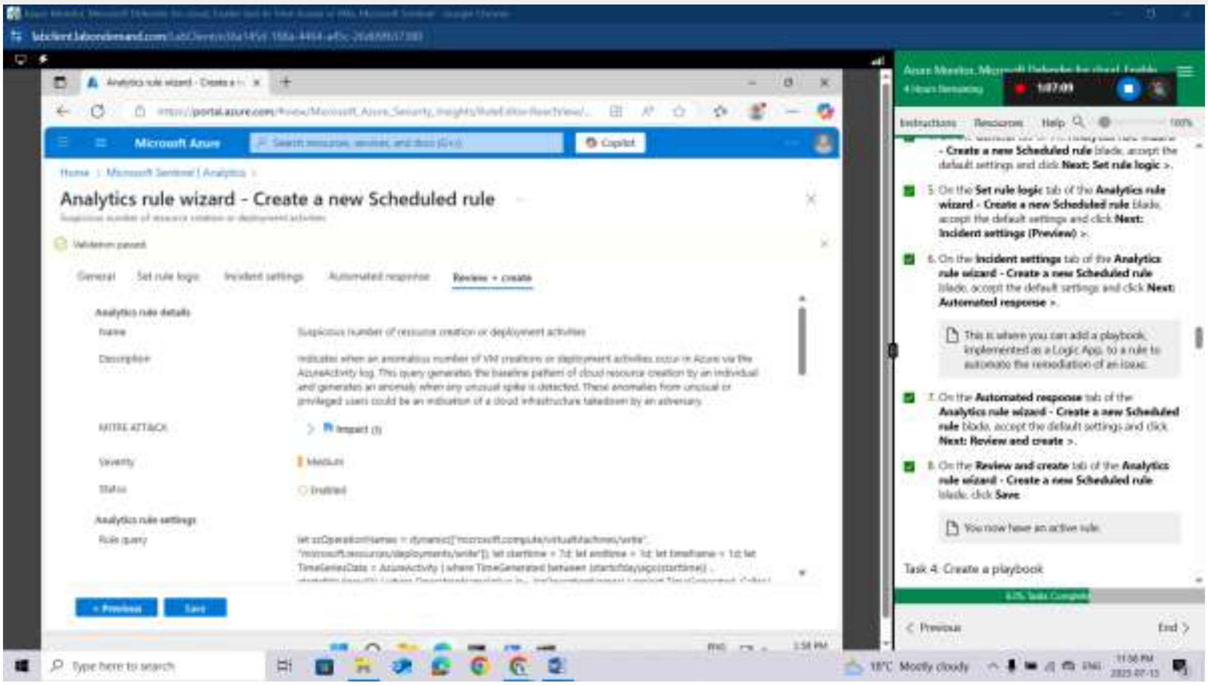
On the **Incident settings** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, accept the default settings and click **Next: Automated response >**.

This is where you can add a playbook, implemented as a Logic App, to a rule to automate the remediation of an issue.

On the **Automated response** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, accept the default settings and click **Next: Review and create >**.

On the **Review and create** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, click **Save**.

You now have an active rule.



Task 4: Create a playbook

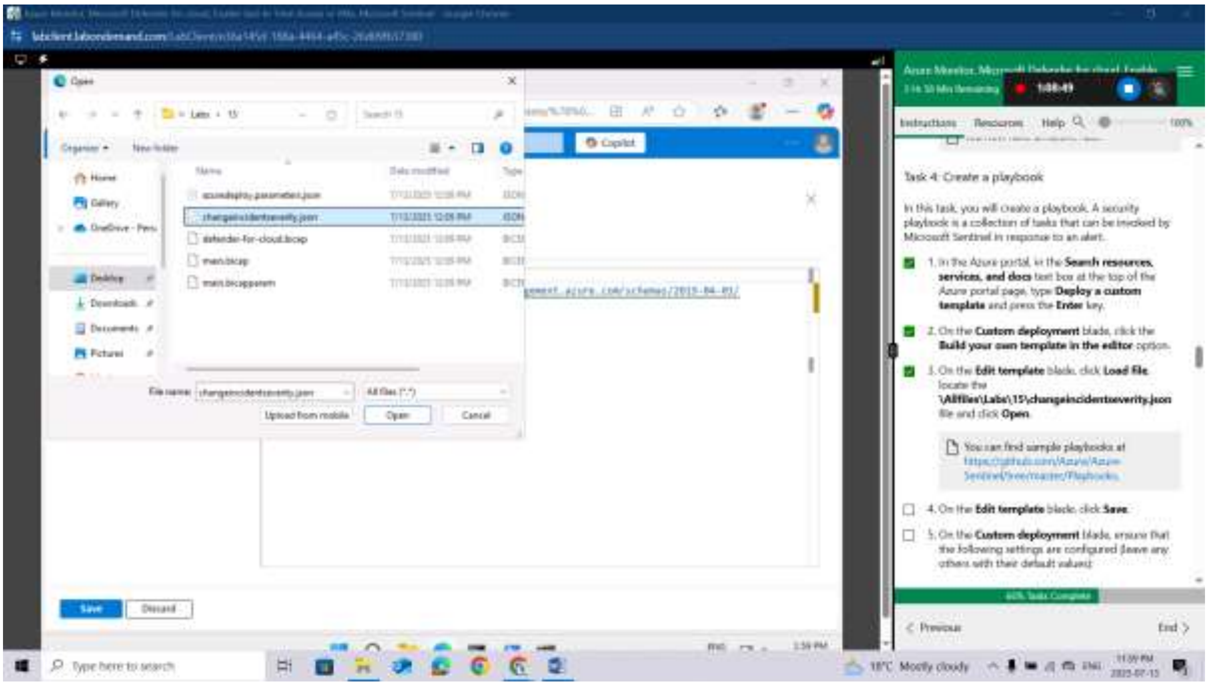
In this task, you will create a playbook. A security playbook is a collection of tasks that can be invoked by Microsoft Sentinel in response to an alert.

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Deploy a custom template** and press the **Enter** key.

On the **Custom deployment** blade, click the **Build your own template in the editor** option.

On the **Edit template** blade, click **Load file**, locate the **\Allfiles\Labs\15\changeincidentseverity.json** file and click **Open**.

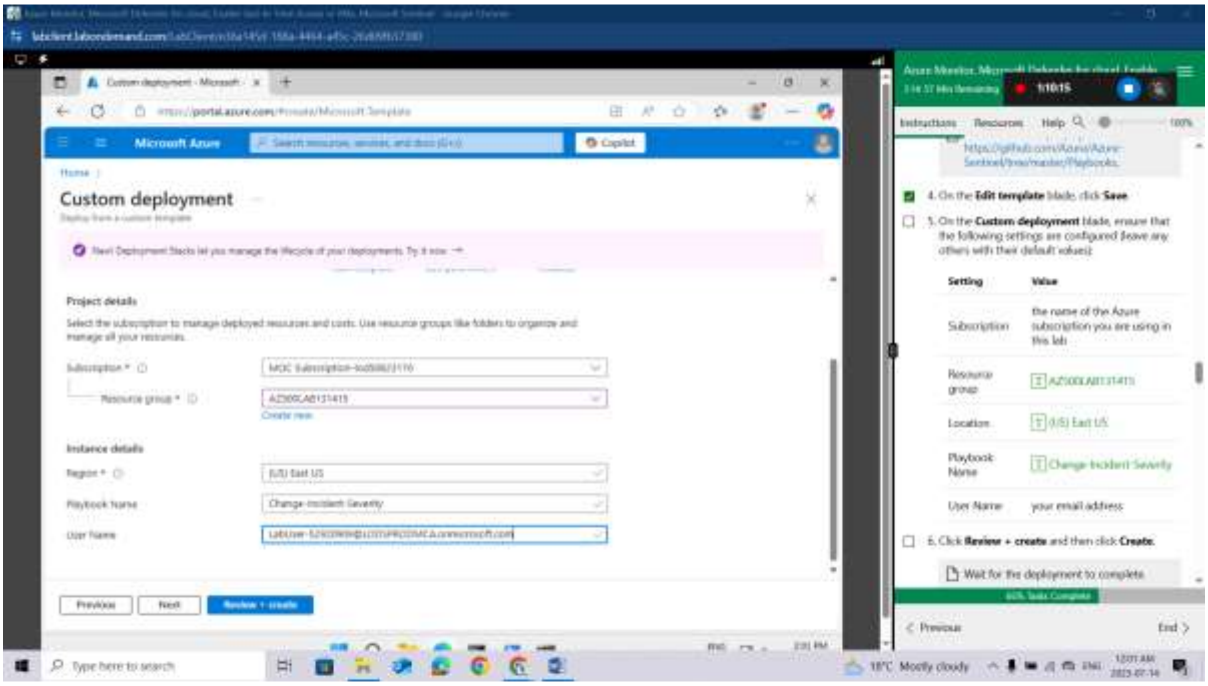
You can find sample playbooks at https://github.com/Azure/Azure-Sentinel/tree/master/Playbooks.



On the **Edit template** blade, click **Save**.

On the **Custom deployment** blade, ensure that the following settings are configured (leave any others with their default values):

Setting	Value
Subscription	the name of the Azure subscription you are using in this lab
Resource group	AZ500LAB131415
Location	(US) East US
Playbook Name	Change-Incident-Severity
User Name	your email address



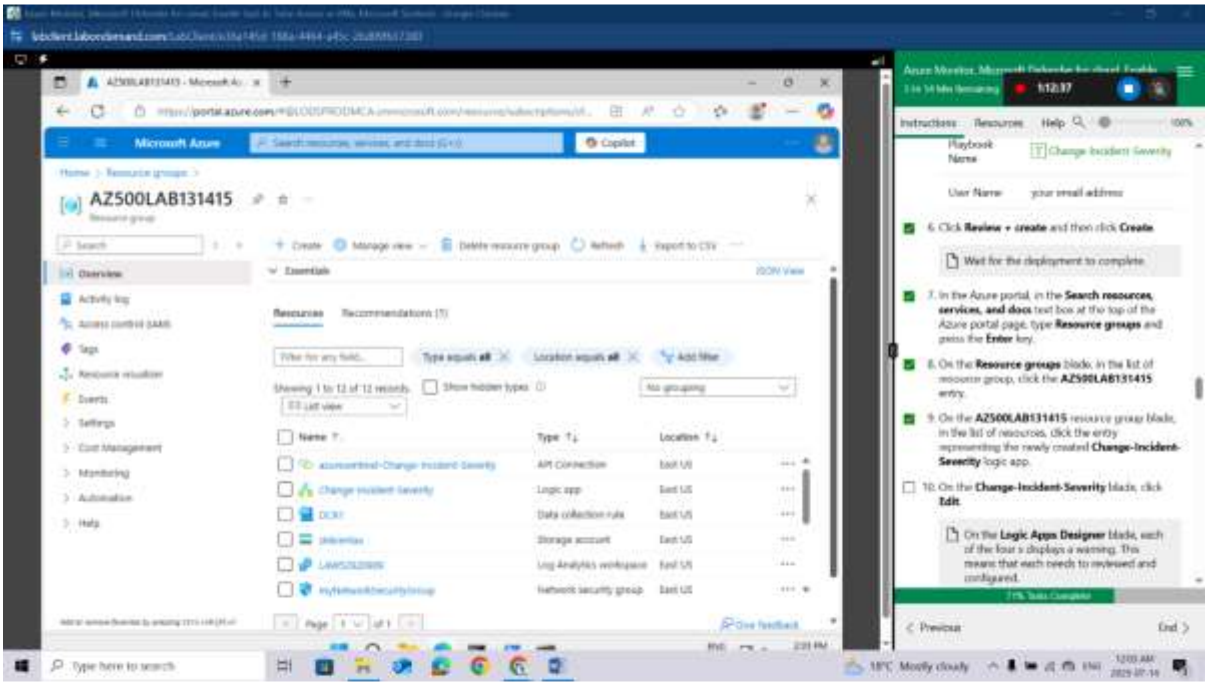
Click **Review + create** and then click **Create**.

Wait for the deployment to complete.

In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Resource groups** and press the **Enter** key.

On the **Resource groups** blade, in the list of resource group, click the **AZ500LAB131415** entry.

On the **AZ500LAB131415** resource group blade, in the list of resources, click the entry representing the newly created **Change-Incident-Severity** logic app.

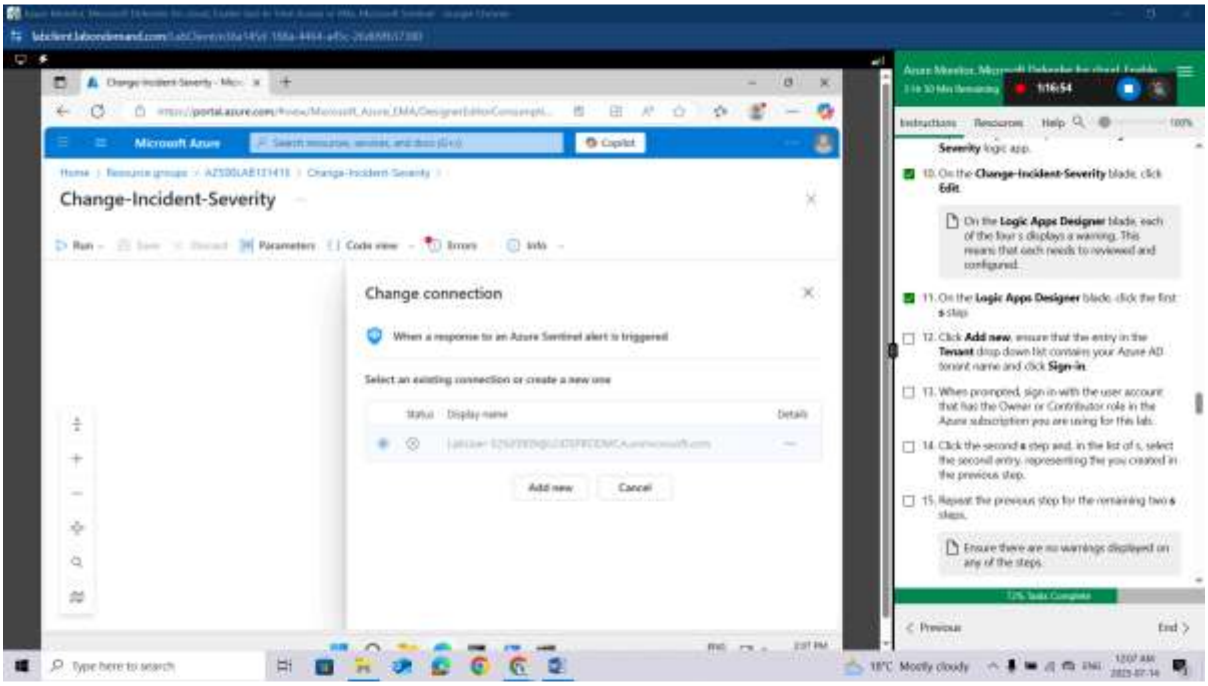


On the **Change-Incident-Severity** blade, click **Edit**.

*On the **Logic Apps Designer** blade, each of the four s displays a warning. This means that each needs to reviewed and configured.*

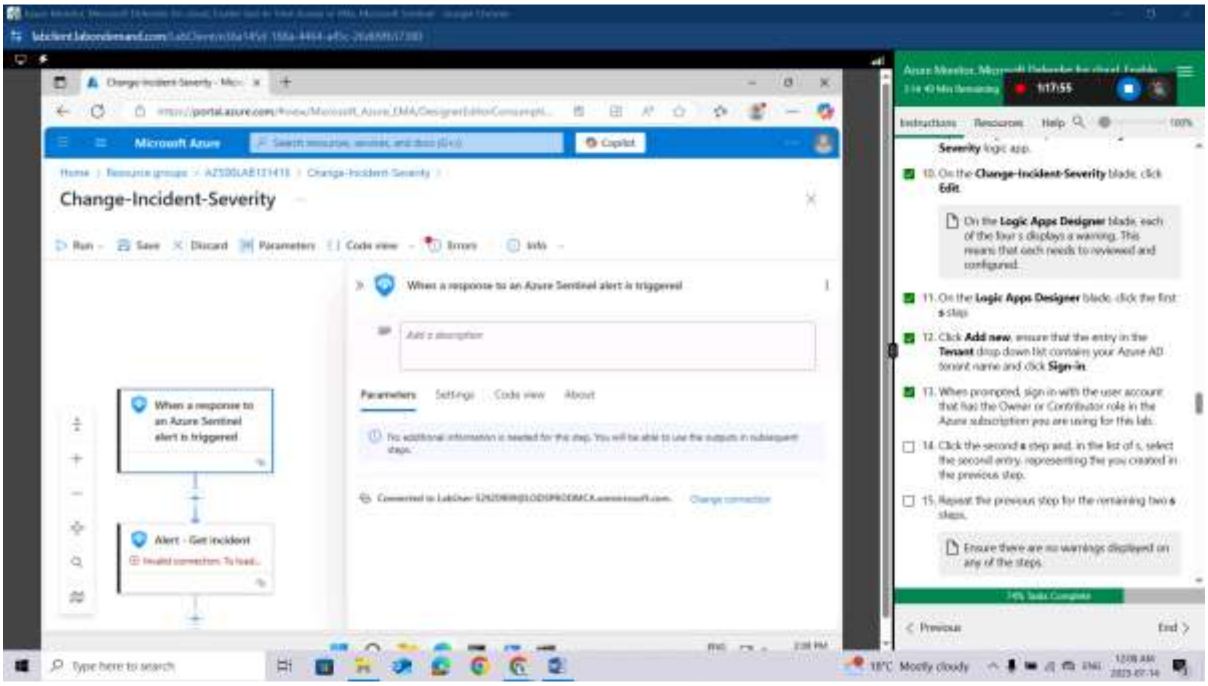
On the **Logic Apps Designer** blade, click the first **s** step.

Click **Add new**,

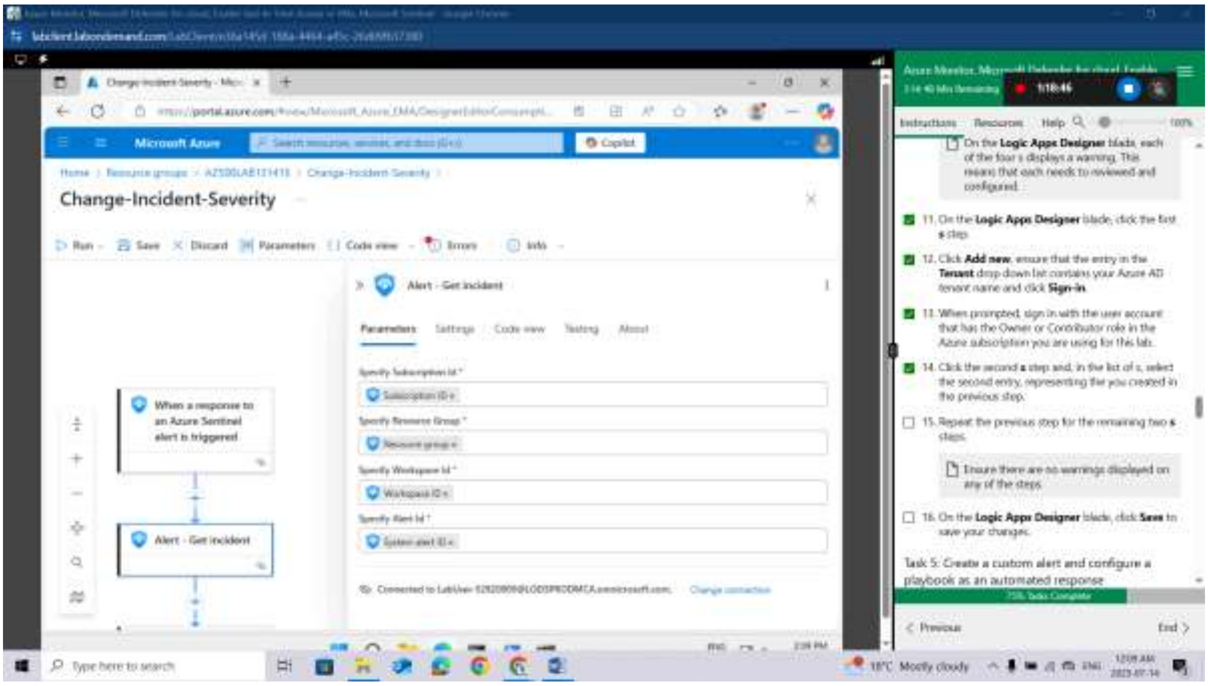


ensure that the entry in the **Tenant** drop down list contains your Azure AD tenant name and click **Sign-in**.

When prompted, sign in with the user account that has the Owner or Contributor role in the Azure subscription you are using for this lab.



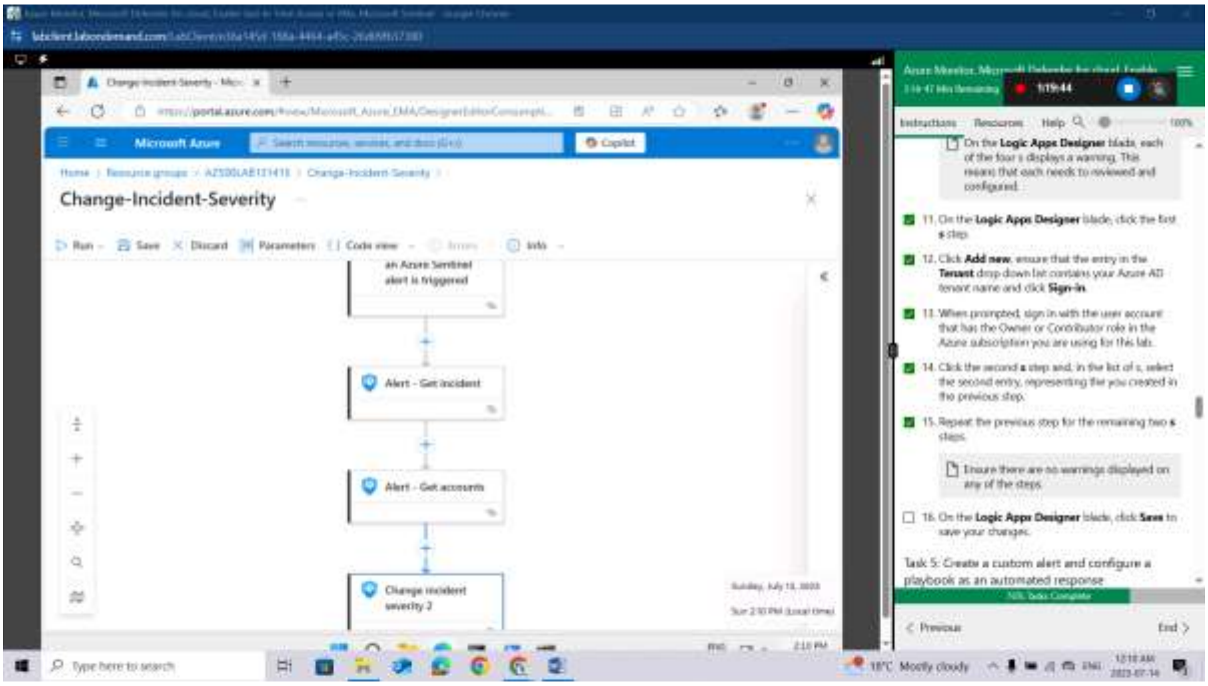
Click the second **s** step and, in the list of s, select the second entry, representing the you created in the previous step.



Repeat the previous step for the remaining two **s** steps.

Ensure there are no warnings displayed on any of the steps.

On the **Logic Apps Designer** blade, click **Save** to save your changes.

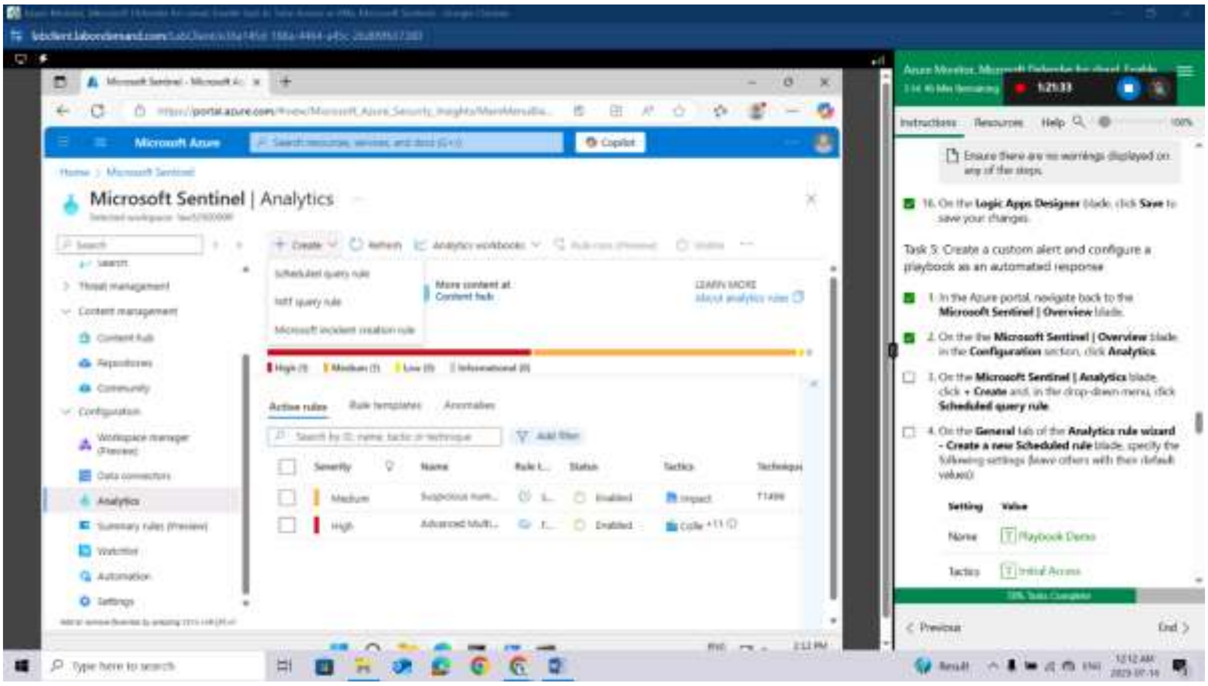


Task 5: Create a custom alert and configure a playbook as an automated response

In the Azure portal, navigate back to the **Microsoft Sentinel | Overview** blade.

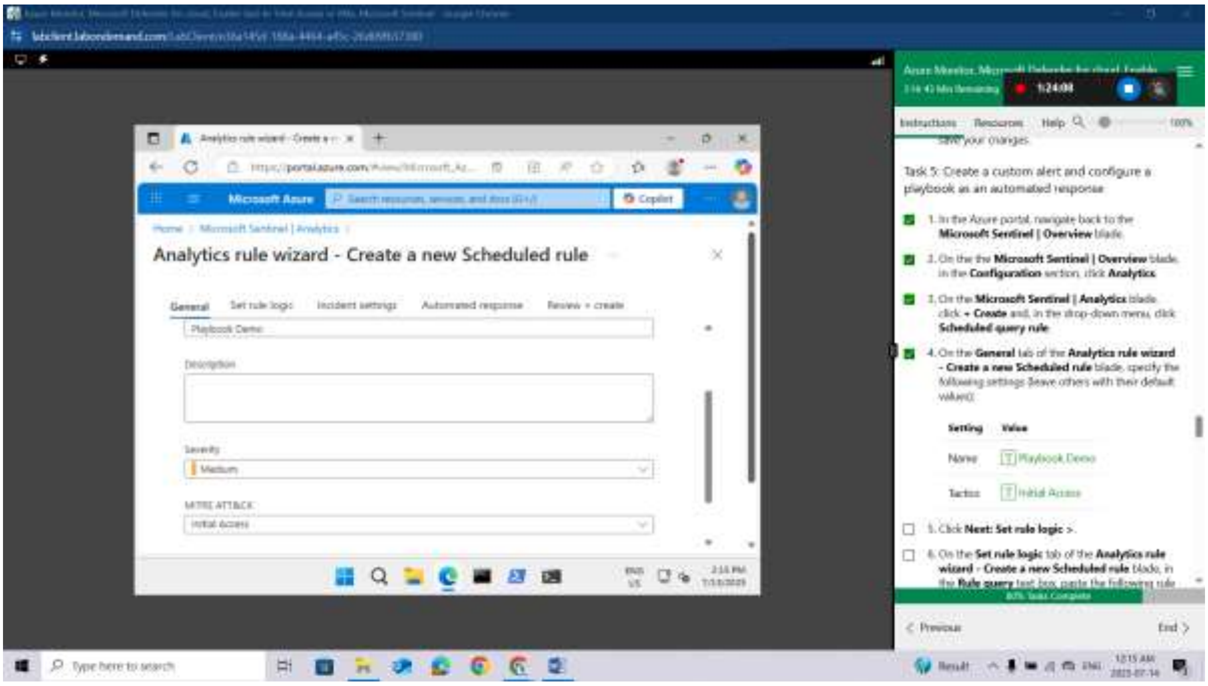
On the the **Microsoft Sentinel | Overview** blade, in the **Configuration** section, click **Analytics**.

On the **Microsoft Sentinel | Analytics** blade, click **+ Create** and, in the drop-down menu, click **Scheduled query rule**.



On the **General** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, specify the following settings (leave others with their default values):

Setting	Value
Name	Playbook Demo
Tactics	Initial Access



Click **Next: Set rule logic >**.

On the **Set rule logic** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, in the **Rule query** text box, paste the following rule query.

```
AzureActivity
 | where ResourceProviderValue =~ "Microsoft.Security"

 | where OperationNameValue =~
"Microsoft.Security/locations/jitNetworkAccessPolicies/delete"
```

This rule identifies removal of Just-in-time VM access policies.
Note *if you receive a parse error, intellisense may have added values to your query. Ensure the query matches otherwise paste the query into notepad and then from notepad to the rule query.*

On the **Set rule logic** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, in the **Query scheduling** section, set the **Run query every** to **5 Minutes**.

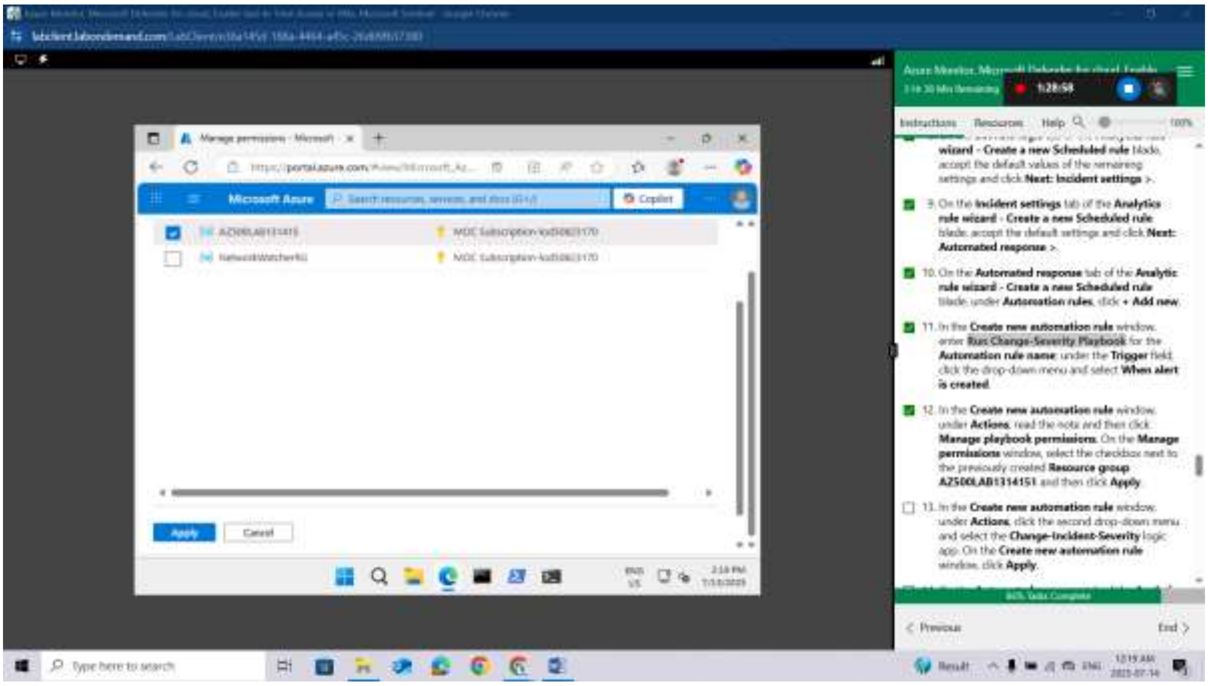
On the **Set rule logic** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, accept the default values of the remaining settings and click **Next: Incident settings >**.

On the **Incident settings** tab of the **Analytics rule wizard - Create a new Scheduled rule** blade, accept the default settings and click **Next: Automated response >**.

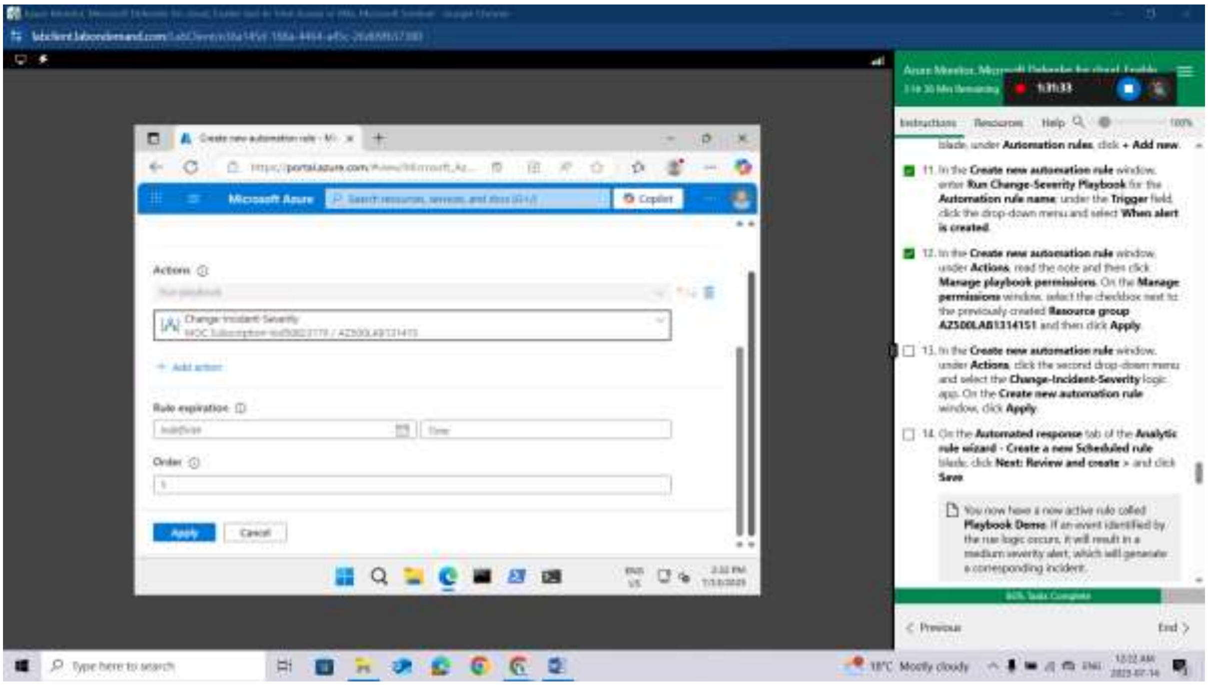
On the **Automated response** tab of the **Analytic rule wizard - Create a new Scheduled rule** blade, under **Automation rules**, click **+ Add new**.

In the **Create new automation rule** window, enter **Run Change-Severity Playbook** for the **Automation rule name**; under the **Trigger** field, click the drop-down menu and select **When alert is created**.

In the **Create new automation rule** window, under **Actions**, read the note and then click **Manage playbook permissions**. On the **Manage permissions** window, select the checkbox next to the previously created **Resource group AZ500LAB1314151** and then click **Apply**.

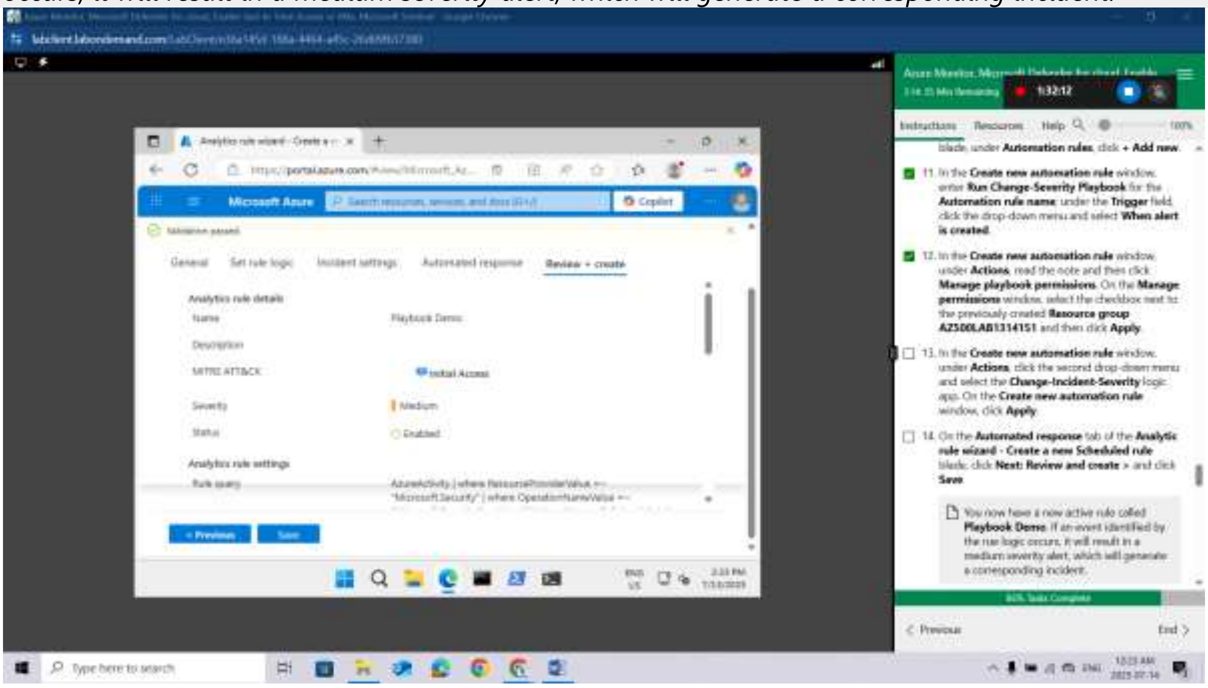


In the **Create new automation rule** window, under **Actions**, click the second drop-down menu and select the **Change-Incident-Severity** logic app. On the **Create new automation rule** window, click **Apply**.



On the **Automated response** tab of the **Analytic rule wizard - Create a new Scheduled rule** blade, click **Next: Review and create >** and click **Save**

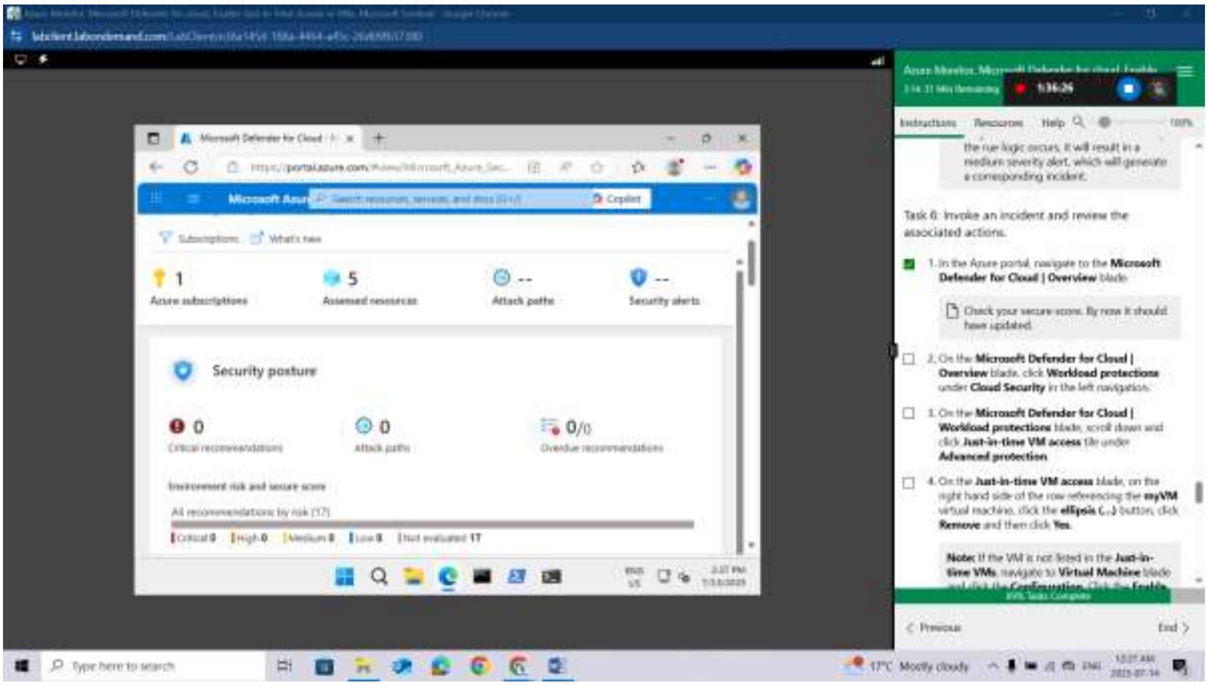
*You now have a new active rule called **Playbook Demo**. If an event identified by the rue logic occurs, it will result in a medium severity alert, which will generate a corresponding incident.*



Task 6: Invoke an incident and review the associated actions.

In the Azure portal, navigate to the **Microsoft Defender for Cloud | Overview** blade.

Check your secure score. By now it should have updated.

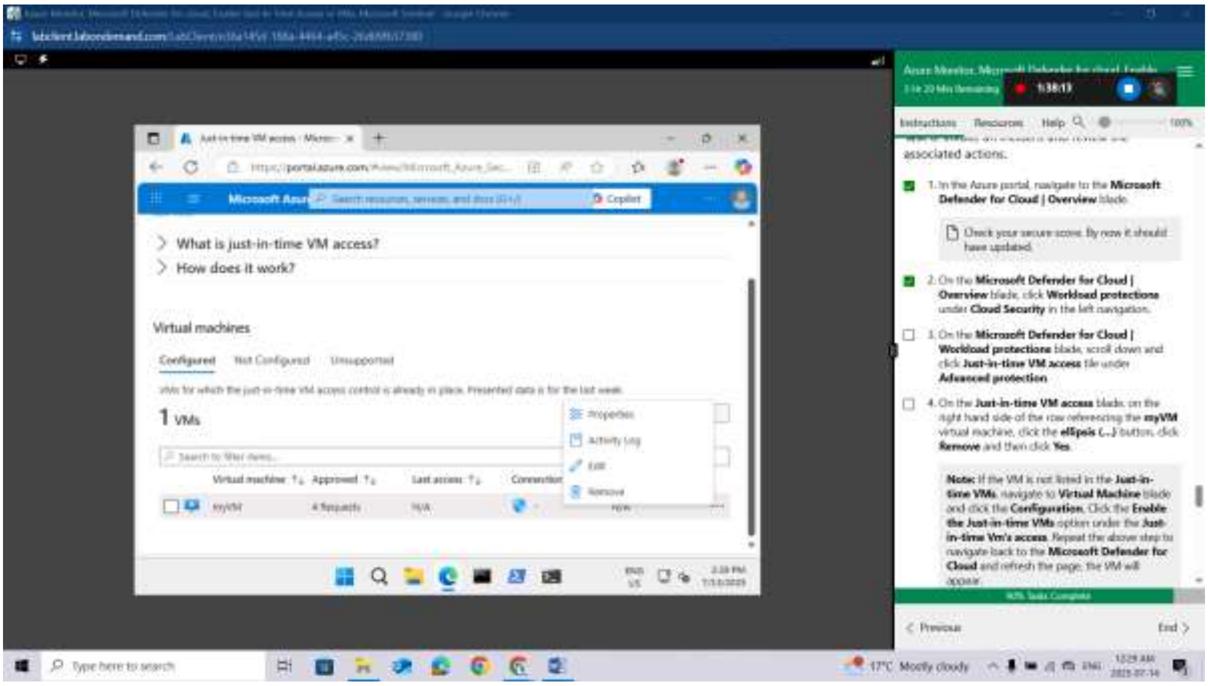


On the **Microsoft Defender for Cloud | Overview** blade, click **Workload protections** under **Cloud Security** in the left navigation.

On the **Microsoft Defender for Cloud | Workload protections** blade, scroll down and click **Just-in-time VM access** tile under **Advanced protection**.

On the **Just-in-time VM access** blade, on the right hand side of the row referencing the **myVM** virtual machine, click the **ellipsis (...)** button, click **Remove** and then click **Yes**.

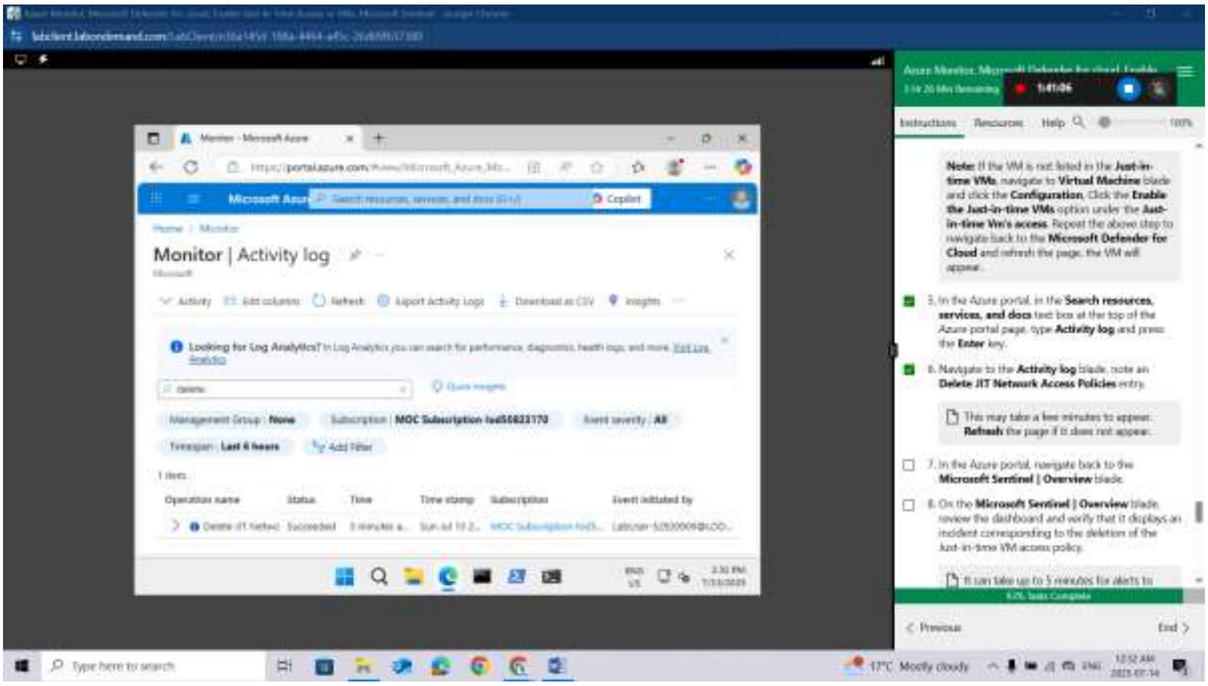
Note: *If the VM is not listed in the* ***Just-in-time VMs****, navigate to* ***Virtual Machine*** *blade and click the* ***Configuration****, Click the* ***Enable the Just-in-time VMs*** *option under the* ***Just-in-time Vm's access****. Repeat the above step to navigate back to the* ***Microsoft Defender for Cloud*** *and refresh the page, the VM will appear.*



In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Activity log** and press the **Enter** key.

Navigate to the **Activity log** blade, note an **Delete JIT Network Access Policies** entry.

This may take a few minutes to appear. ***Refresh*** *the page if it does not appear.*



In the Azure portal, navigate back to the **Microsoft Sentinel | Overview** blade.

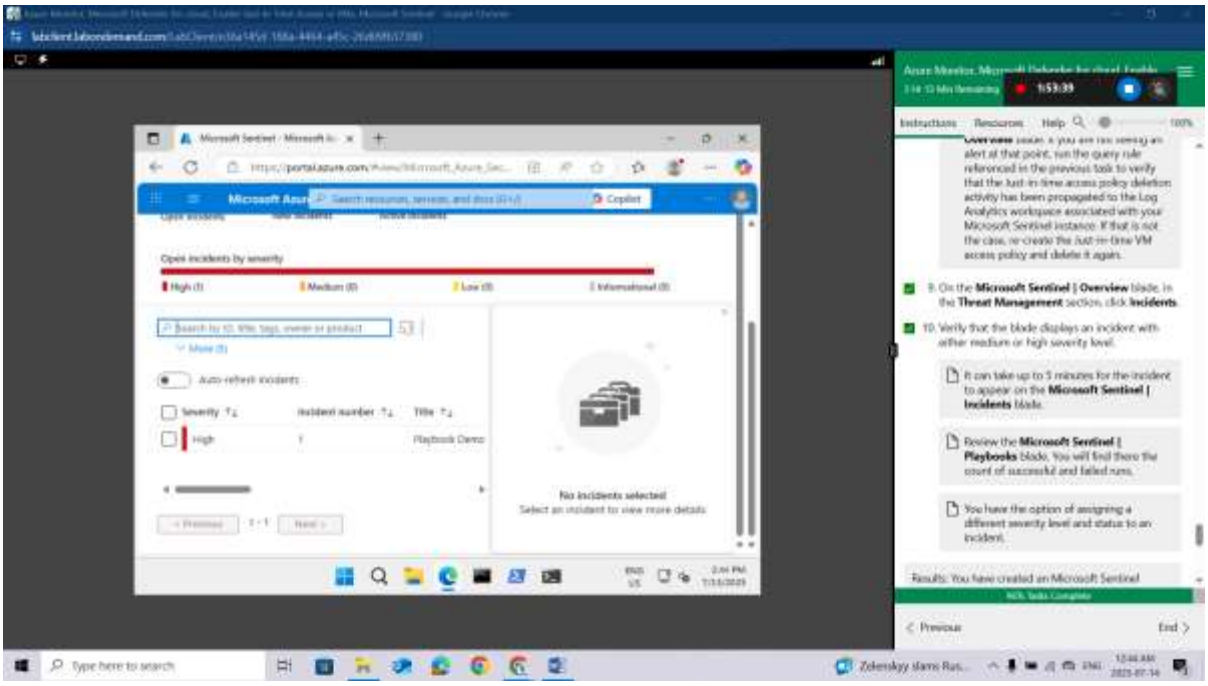
On the **Microsoft Sentinel | Overview** blade, review the dashboard and verify that it displays an incident corresponding to the deletion of the Just-in-time VM access policy.

*It can take up to 5 minutes for alerts to appear on the **Microsoft Sentinel | Overview** blade. If you are not seeing an alert at that point, run the query rule referenced in the previous task to verify that the Just-in-time access policy deletion activity has been propagated to the Log Analytics workspace associated with your Microsoft Sentinel instance. If that is not the case, re-create the Just-in-time VM access policy and delete it again.*

On the **Microsoft Sentinel | Overview** blade, in the **Threat Management** section, click **Incidents**.

Verify that the blade displays an incident with either medium or high severity level.

*It can take up to 5 minutes for the incident to appear on the **Microsoft Sentinel | Incidents** blade. Review the **Microsoft Sentinel | Playbooks** blade. You will find there the count of successful and failed runs.*
You have the option of assigning a different severity level and status to an incident
Results: You have created an Microsoft Sentinel workspace, connected it to Azure Activity logs, created a playbook and custom alerts that are triggered in response to the removal of Just-in-time VM access policies, and verified that the configuration is valid.



Clean up resources

Remember to remove any newly created Azure resources that you no longer use. Removing unused resources ensures you will not incur unexpected costs.

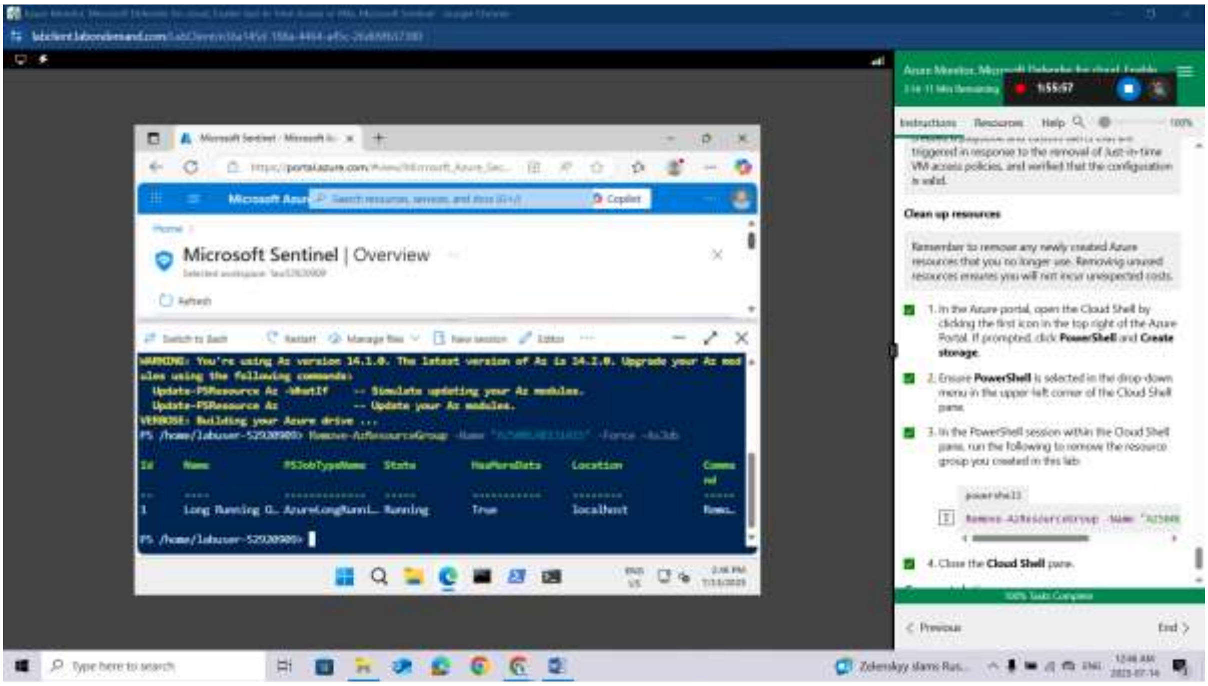
In the Azure portal, open the Cloud Shell by clicking the first icon in the top right of the Azure Portal. If prompted, click **PowerShell** and **Create storage**.

Ensure **PowerShell** is selected in the drop-down menu in the upper-left corner of the Cloud Shell pane.

In the PowerShell session within the Cloud Shell pane, run the following to remove the resource group you created in this lab:

```powershell
Remove-AzResourceGroup -Name "AZ500LAB131415" -Force -AsJob
```

Close the **Cloud Shell** pane.



Congratulations!You have successfully completed this lab. Click **End** to mark the lab as **Complete**.

Conclusion

In this lab, I successfully configured and utilized multiple Azure security and monitoring services. I used **Azure Monitor** to track system performance and gain operational insights. Through **Microsoft Defender for Cloud**, I assessed the security posture of my resources and applied proactive recommendations. I implemented **Just-In-Time (JIT) access** to secure VM endpoints and minimize unnecessary exposure. Finally, I explored **Microsoft Sentinel** to detect threats and enable effective incident response using automation and analytics. Overall, the lab reinforced the importance of integrating these services to achieve a secure and well-monitored Azure environment.